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A global network for energy-efficient shipping

MTCC CARIBBEAN
Maritime Technology Cooperation Centre

MARITIME TECHNOLOGY COOPERATION CENTRE CARIBBEAN (MTCC Caribbean)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY

THE GLOBAL MTCC NETWORK (GMN) PROJECT

Technology Needs Assessment (TNA)

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1 Summary

Recognizing that the maritime transport sector is responsible for a sizeable contribution to global CO_2 emissions, strides have been made toward reducing such emissions from vessels, in addition to pursuing reductions in other harmful greenhouse gas emissions. These developments include international legal instruments as well as advancements in Energy Efficient (EE) and Emission Abatement Technologies and methods. However, gaps do exist in the wake of these developments as the uptake of new technologies, in particular, is not uniform globally.

This Technology Needs Assessment (TNA) focuses on the technological aspect of these developments in the context of the Caribbean. Five categories of stakeholders within the Caribbean maritime cluster were identified and targeted for this assessment, namely; maritime administrations, owners/operators, shipyards/dockyards, technology providers, and ports. A couple of the maritime administrations indicated that a technology needs assessment has already been carried out in their country, but the majority indicated that none has been undertaken thus far. Survey instruments were drafted and disseminated to the stakeholders and the results analyzed to determine the technology needs for the Caribbean region.

The results of the analysis indicated both the demand and supply for existing EE and emission abatement technologies within the Caribbean region, with disparities being identified between owners' and operators' awareness of what is available, and what is actually available. Demand for EE and emission abatement technology is affected by the level of awareness of those technologies, and the awareness level of all stakeholders was determined, with owners/operators being the most aware. On the supply side, stakeholders identified gaps of varying magnitudes between technology available within the Caribbean and internationally available technologies, indicating that the supply within the region lags behind what is globally available.

Several factors affect the decisions of stakeholders with respect to their technological needs and selection of appropriate technology. The stakeholders, using a predetermined list of technology by MTCC Caribbean, ranked the factors that affect their decision making process. It was determined that the most important factors are financial viability and regulatory compliance.

They also identified the barriers that impact the implementation of maritime technology transfer. The level of impact of each barrier was rated to determine the greatest challenges faced by the region regarding the uptake of EE and emission abatement technology. The largest barriers were determined to be high investment costs, lack of finance and the fact that in some instances there are no regulatory requirements.

The assessment determined that in order to overcome these barriers, an enabling environment has to be created which will include a robust and comprehensive regulatory framework, improvements in policy, the removal of organizational and market barriers, and the provision of funding mechanisms, among others. To facilitate the creation of an enabling environment, a technology action plan was developed which included proposed solutions to each of the identified barriers; as means of facilitating the uptake of technology in the region.

2 Overview

Maritime transport emits around 1000 million tonnes of CO_2 annually and is responsible for about 2.5% of global Greenhouse Gas (GHG) emissions (3rd International Maritime Organization (IMO) GHG study). Shipping emissions are predicted to increase between 50% and 250% by 2050 – depending on future economic and energy developments. Ships' energy consumption and CO_2 emissions could be reduced by up to 75% by applying operational measures and implementing existing technologies (2nd IMO GHG study).

In July 2011, the IMO adopted mandatory measures to improve the EE of international shipping through resolution MEPC.203(62). This was the first legally binding instrument to be adopted since the Kyoto Protocol that addresses GHG emissions and the first global mandatory GHG-reduction regime for an international industry sector.

All ships of 400 GT and above, engaged in international trade, are required to implement and maintain a Ship Energy Efficiency Management Plan (SEEMP) that establishes a mechanism for operators to improve the energy efficiency of ships. This should be achieved by monitoring the energy efficiency performance of a ship's transportation work, using, for example, the Energy Efficiency Operational Indicator (EEOI) as a monitoring and/or benchmarking tool and, at regular intervals, considering new technologies and practices to improve energy efficiency.

The IMO's MEPC adopted the Initial IMO GHG Strategy in April 2018 (MEPC 72) which marked another significant milestone in addressing maritime shipping emissions. The initial strategy identifies levels of ambition for the international shipping sector noting that technological innovation and the global introduction of alternative fuels and/or energy sources for international shipping will be integral to achieving the overall ambition. Under the short term measures, the initial strategy includes measures to support innovative technologies to further enhance the energy efficiency of ships and incentives for first adopters to develop and take up new technologies. It is recognizing that technological developments would enable significant improvements of the current standards and reduce the harmful impact that ships' exhaust gases may have on the environment, ecosystems and human health.

The strategy also identifies the special needs that LDCs and SIDS have in particular with regard to capacity building and technical cooperation. Along with assisting in the efforts to promote low-carbon technologies by facilitating public-private partnerships and information exchange, the IMO continues to provide mechanisms for facilitating information sharing, technology transfer, capacity-building and technical cooperation, taking into account resolution MEPC.229(65) on the Promotion of technical co-operation and transfer of technology relating to the improvement of energy efficiency of ships.

Many of these measures are cost-effective and offer net benefits, as reduced fuel costs ensure the pay-back of any operational or investment costs. Further reductions could be achieved by implementing new innovative technologies.

The viability of many technological options for reducing emissions from ships is becoming better established as a result of a range of initiatives at international and national levels. Nevertheless, there is a need to find ways by which developing countries can formalize the important process of assessing the emerging technologies and practices for adapting to climate change and mitigating greenhouse gas emissions from Maritime sector.

Realizing the decarbonisation targets of the sectors is dependent on the adoption of emissions control and abatement technologies and measures. However, continuous and widespread deployment of negative emissions technologies is currently not happening at the required scale.

A TNA involves the identification and evaluation of technical means for achieving specified ends. From a climate change and emission reduction perspective, the TNA indicates the priorities of stakeholders in the adoption of technologies, practices, and policy reforms that can be implemented on ships to reduce GHG emissions and/or to adapt to the impacts of climate change by enhancing resilience and/or contributing to sustainable development goals.

3 Key Technologies

The maritime industry has pursued a number of technical and operational measures for reducing GHG Emissions from ships at sea and at berth. Technical measures include Emissions Control and Energy Efficiency Measures (ECEEMs); and operational measures include economies of scale, improved voyage planning and the use of alternative fuels.

The possible emission and energy consumption reduction measures therefore can be achieved by using various technical and operational measures called Energy Saving Potentials (ESPs). These ESPs can be broadly divided into three categories;

- Hull resistance reduction,
- Propulsion efficiency improvement, and
- Reduction in energy demand.

For emission reduction, technological measures can further be divided into two major categories;

- Alternative fuels, and
- Renewable sources (Alternative sources of energy)

Bouman et al., 2017 reviewed approximately 150 studies to determine the CO_2 reducing potential of some key technologies available to the industry. The high degree of uncertainty in CO_2 emission reduction potential for various ECEEMs was highlighted in these studies. The reason for this uncertainty was due to specific type of vessels and different model assumptions used in these studies. Table 1 provides a summary of the findings.

Table 1 - CO_2 emissions reduction potential for various ECEEMs (Bouman et al., 2017)

Type of measure	Measure reviewed	Potential CO_2 Reduction range (%)	
		Lowest Potential	Highest Potential
HULL DESIGNS AND MODIFICATIONS	Vessel Size	4.0%	83.0%
	Hull Shape	2.0%	30.0%
	Lightweight materials	0.1%	22.0%
	Air Lubrication	1.0%	15.0%
	Resistance reduction devices	2.0%	15.0%
	Ballast water reduction	0.0%	10.0%
	Hull Coating	1.0%	10.0%
POWER AND PROPULSION SYSTEMS	Hybrid power / propulsion (Hybrid electric)	2.0%	45.0%
	Power system / machinery	1.0%	35.0%
	Propulsion efficiency devices	1.0%	25.0%

	Waste heat recovery	1.0%	20.0%
	Reduced On board power demand (lighting etc.)	0.1%	3.0%
ALTERNATIVE FUELS	Biofuels	25.0%	84.0%
	LNG	5.0%	30.0%
ALTERNATIVE ENERGY / RENEWABLE ENERGIES	Wind power / kite, sail, wings	1.0%	50.0%
	Fuel Cells	2.0%	20.0%
	Cold Ironing / shore power	3.0%	10.0%
	Solar Power / solar panels	0.2%	12.0%
OPERATIONAL MEASURES	Speed Optimization / speed reduction / optimization	1.0%	60.0%
	Capacity utilization / vessel and fleet management	5.0%	50.0%
	Voyage optimization / weather routing, route planning	0.1%	48.0%
	Other measures / Trim and draft optimization/ energy management	1.0%	10.0%

3.1 Resistance Reduction Methodologies

The effect of water resistance on the movement of vessels is associated with energy loss, and is a major factor in fuel consumption. The total resistance can be divided into; resistance due to viscous; wave making & wave breaking; air resistance; and appendage resistance. In the design phase “length to beam ratio” and beam to draft ratio along with other dimensions also affect ship resistance. The selection of optimal dimensions i.e. global optimization, also reduces resistance.

3.1.1 Frictional Resistance Reduction

Speed reduction – The ship resistance increases non-linearly with increase in speed. Therefore, any reduction in speed leads to reduction in resistance.

Paints – The advancements in paint technology has extended the role of hull coating from reducing fouling to reducing frictional drag. Hull condition and coating is vital as a 1% increase in power is required to maintain speed for a 10 micro millimeter increase in hull roughness.

Air Lubrication – In this viscous resistance reduction measure, an underwater air surface is provided. The three basic air lubrication methods are:

- **Air cavity** – Underwater hull is arranged in a way to allow recirculation of air. Such air pockets reduce the viscous resistance by reducing the contact between hull and water.
- **Micro Bubbles** – Small bubbles pumped into the underwater hull surface close to the boundary layer of the ship.
- **Air Film** – Use of special paints to attract air to the hull surface as a film, however the film formation is influenced by the water flow speed across the hull.

3.1.2 Hull Form and Appendage optimization.

Bulbous bow – The use of a Bulbous bow creates a wave interaction which results in reduced wave resistance. The benefit of a bulbous bow increases with an increase in speed. It is also more effective on the full load condition compared to draught in higher conditions.

Stern form – Resistance reduction up to 5% is possible by stern modifications.

Bilge keel – The bilge keel provides additional longitudinal strength with roll- stability improvement. Ideally it should not add to the resistance and therefore requires advanced Computational Fluid Dynamics (CFD) optimizations.

3.1.3 Superstructure optimization

Minimizing the superstructure with minimal drag shapes reduces air resistance. Air resistance is directly dependent on the transverse cross sectional area of a vessel above the water line. The appendage modifications not only affect resistance but also the propulsion efficiency.

3.2 Propeller Design and Efficiency

Propeller efficiency is defined by the propeller geometry. Propeller efficiency is highly affected by cavitation. However, changes in blade ratio, geometry, skewness and blade section prevent or delay cavitation. A number of active and passive devices in addition to hull form improve wake fraction, thrust, rotative and open water efficiency.

Stern hull – form and positioning of the propeller.

Wake improvement and Separation Avoidance Devices – These devices optimize wake to propeller disk. Examples include Grothues spoilers, Schneekluth ducts and Mevis duct stern tunnels.

Pre-swirl devices – fins and stators placed in front of the propeller to generate rotational flow in front of the propeller. Stator duct combination is usually recommended on bulk carriers.

Post-swirl devices – These devices convert rotational flow to axial flow and hence placed behind the propeller. This also improves rudder performance, for example - boss cap fin.

3.3 Reduction in energy demand

Shaft generator – Shaft generator can be utilized if controllable pitch propeller is used. Fixed pitch propeller shafts can also be used but only during sea voyages. Shaft generator increases the power demand on main engine but saves on the fuel consumption for electrical power generator.

Waste heat Recovery – the exhaust gas from the main engine can be used to generate steam. The surplus heat is utilized to drive the exhaust gas turbine or to fulfil the other requirements on the ship. The heat recovery can be made through conventional heat exchangers from cooling water and recovery from exhaust gas boiler (EGB). However, this is only available when the main engine is running.

Weight optimization – The displacement of the vessel not only increases resistance but also demands higher propulsion power. Weight savings can be achieved using a variety of measures. For example, the use of high tensile steel can offer up to 2% reduction in fuel consumption.

Efficient Lighting system – Installing a low energy consumption system will reduce the power requirements of auxiliary engines on-board. It will also reduce the energy demands for the lighting of the ship.

Use of Renewable Energy – Kite sailing; very large free flying sails, supplement the propulsion generated by the engines and reduce fuel consumption by using wind power. The system consists of a kite, an automotive control system and a launch & recovery system.

Table 2 below presents the summary of various methods:

Table 2 - Summary of various measures

Reduction Measure	**Various Methods**
Hull Resistance	Hull Shape Optimal Dimensions B/T ratio Use of coatings Bow and Stern optimisation Minimising Openings on the hull Application of appendages
Propulsion System	Selection of constant speed lower power engine
Propeller Selection	Geometry of propeller Diameter, number of blades Operating RPM Effective draft Use of various optimisation techniques
Machinery System	Attained EEDI is directly based on the MCR Application of electronic controls Reduction in specific fuel consumption by improving machinery parts
Auxiliary Systems	Number, Size and connection Use of shaft generator Utilising motors

Reduction Measure	**Various Methods**
	Lighting system on board ship
Ship Structure	Light weight material in construction Increased use of High tensile steel and aluminium Reduction in displacement of the vessel reduces the power requirement
Innovative Measures	Air cavity Micro Bubbles lubrication Hull pattern and texture Use of renewable sources

In addition to measures being implemented by ships, major ports around the world have already begun implementing programmes and policies to address emissions from ships while in or arriving at port. These programmes are a mix of operational, technical, regulatory and incentive measures. On the technical and operational front, measures such as cold ironing and just in time arrival are being supported and encouraged. Port incentive programmes that favour and attract ships with modern and GHG efficient designs could have some effect (Winnes et al., 2015). However, greater impacts are more likely to result from international regulations than from individual port programmes.

Single measures or incentives will not provide significant reductions in GHG emissions but a combination of a number of them have proven to be most beneficial. According to the IMO, it has been estimated that through ESPs, ships' energy consumption and emissions can be reduced by as much as 75% by implementing existing technologies and proven operational measures. However applying these solutions with such high degrees of performance uncertainties and costly investments has resulted in sluggish industry action; this TNA aims to, among others, determine the technology needs of the region to improve the sluggishness of industry response.

4 Methodology

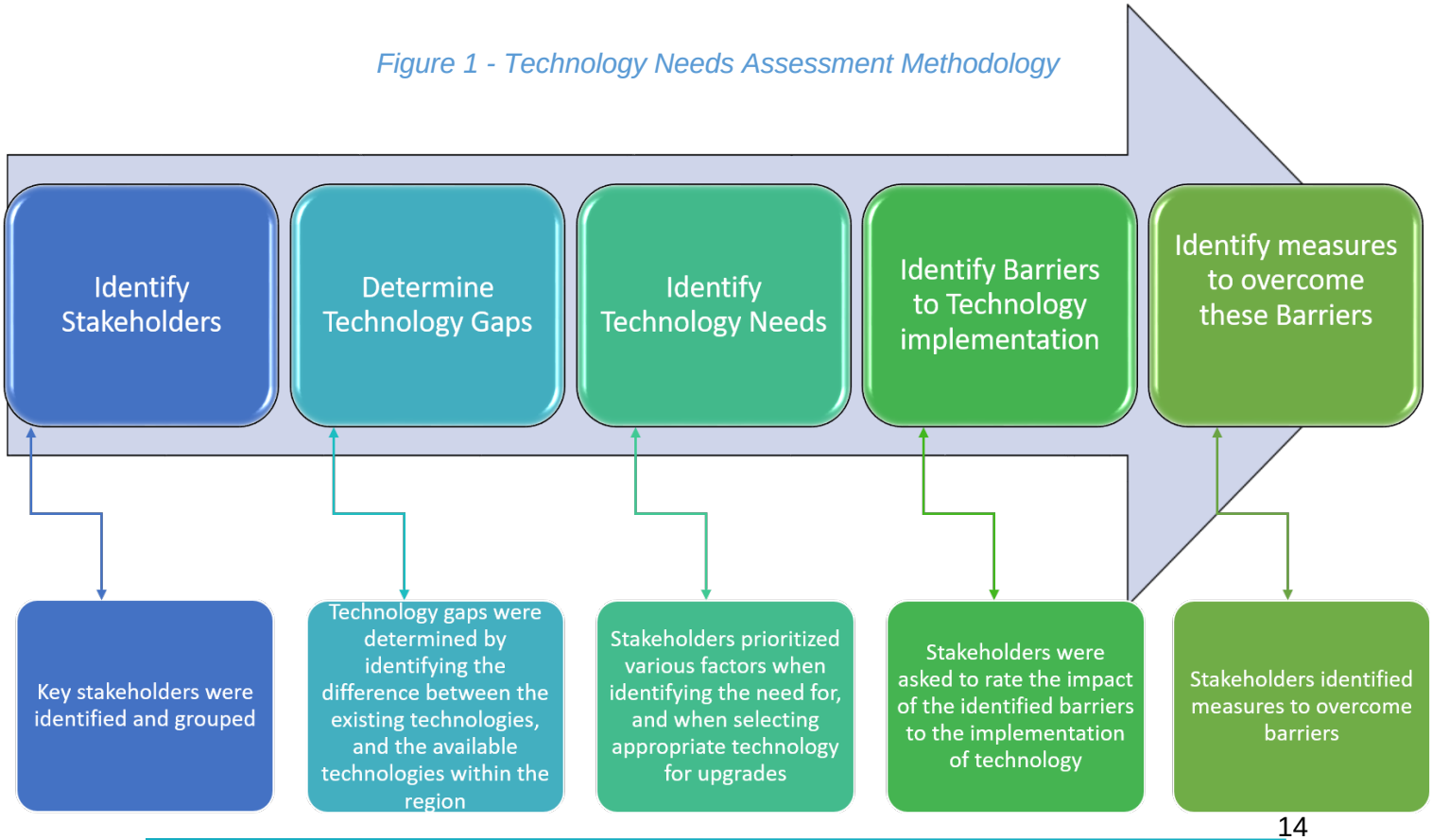
Available research and literature in shipping indicate that it is possible to improve energy efficiency and reduce emissions in a cost-effective manner using technology (Lindstad et al., 2015a; Buhaug et al., 2009; Faber et al., 2009, 2011; Alvik et al., 2010). However, these studies are limited in scope as they address mainly CO_2 reduction potential at the macro level and/or investigation of individual or combined technological and operational measures reducing emissions.

There is an abundance of studies presenting mitigation measures and scenarios for shipping but comprehensive reviews on this issue are limited in scientific literature. To quantify the maximum potential for GHG reduction and energy efficiency measures is complex due to various organizational, technical, economic and political barriers for implementation. The application of these measures is also dependent on distinct ship types and age. Therefore, it is essential to take into account the viewpoint of each stakeholder. This TNA attempts to capture the viewpoint of stakeholders within the Caribbean Region. In this regard, objectives of this technology needs assessment carried out by MTCC Caribbean are to:

1. Determine the available technologies in the participating countries;
2. Determine the technology gaps that exist;
3. Determine the technology needs of participating countries;
4. Determine the barriers to implementing their technology needs; and
5. Determine the actions required to help overcome these barriers.

The TNA was undertaken in accordance with the steps outlined in figure 1 below:

Figure 1 - Technology Needs Assessment Methodology



4.1 Key Stakeholders

Key stakeholders were identified and grouped according to the following categories:

1. **Maritime Administrations** – These are responsible for regulation and enforcement in their capacities as Flag State Administrations and Port State Control (PSC), which makes them one of the major drivers of technology uptake in the region. Maritime administrations are also the main focal points between the IMO and the local maritime industry, ensuring compliance with international maritime instruments via enforcement through national instruments. As the primary government entity responsible for regulating shipping, maritime administrations can implement incentive programmes that will encourage vessels calling within their jurisdiction to uptake EE and emission abatement technology, cleaner marine fuels, and other methods that achieve reductions in harmful GHG emissions while at port. MTCC Caribbean sought the knowledge and expertise of the maritime administrations of the 16 Caribbean territories under its ambit to determine the needs of the region from the standpoint of the regulators. Refer to appendix I for the list of maritime administrations that provided responses for the technology needs assessment.

2. **Vessel Owners/Operators** – These are the principle users of EE and emission abatement technology. As the end users and those chiefly affected by changes in technology requirements, technology design must balance their needs while meeting international regulatory requirements; making the needs of vessel owners/operators of paramount importance. There have been mixed responses by owners/operators regarding the current and impending 2020 fuel Sulphur content cap. The demand by ship owners/operators for particular technologies is impacted by the availability of that technology; making the determination of their demand of critical importance for the suppliers in the region. They will also be able to identify the barriers and challenges that need to be addressed to achieve compliance with upcoming regulatory requirements. Refer to appendix II for the list of vessel owners/operators that provided responses for the TNA.

3. **Shipyards and Dockyards** – These are the facilitators of green technology uptake. They function as the bridge between technology providers and vessel owners/operators. Shipyards and Dockyards are uniquely positioned to be highly familiar with a wide range of technologies and ship types, and may possess an enhanced understanding of the needs of all parties involved. Their importance as a respondent lies with their ability to provide the supply of and facilitate installation/retrofitting of EE and emission abatement technology. MTCC Caribbean was unable to secure a response from a shipyard/dockyard. Refer to appendix III for a letter from Caribbean Dockyard and Engineering Services Limited (Caridoc) explaining why they believe the survey was not applicable to them.

4. **Technology Providers** – These are the architects of the technology that will be installed, used and regulated by other stakeholders in the maritime cluster. The technology uptake

process is a continuous loop and technology providers are responsible for the incorporation of improvements recommended by the other stakeholders, among others. As a result, they are tasked with balancing the needs of regulators and the needs of vessel owners/operators. The importance of these providers also lies with the availability of their technology to be installed/retrofitted on vessels within the region. Refer to appendix IV for the list of technology providers that provided responses for the technology needs assessment.

5. **Ports** – These entities are multifunctional in nature, as they can undertake roles as both regulators and enablers; the latter referring to the provision of onshore power to reduce vessel emissions while berthed. As such, their input is valuable in determining the technology needs for the region. Similarly to maritime administrations, some ports and port authorities are vested with the authority to establish incentive programmes that will encourage vessels calling at their ports to uptake EE and emission abatement technology, cleaner marine fuels, and other methods that achieve reductions in harmful GHG emissions while at port. These incentives can also be complemented by ports' supply of onshore power which allows vessels to continue to operate while berthed without the use of their engines or generators, which reduces fuel consumption and emissions while alongside. Refer to appendix V for the list of ports that provided responses for the technology needs assessment. Questionnaires specific to the various categories of stakeholders were developed, to:

   I. Determine the demand of the various EE technologies and emission abatement methods;
   II. Identify the barriers to the implementation of these EE technologies and emission abatement methods;
   III. Identify measures to overcome these barriers.

4.2 Determination of Technology Gaps

Existing EE technologies and emission abatement methods were reviewed and documented. Technology gaps were determined by identifying the difference between the existing technologies, and the available technologies within the region.

4.3 Identification of Technology Needs

Technology needs were determined using a two-step approach where respondents were required to prioritize the factors.

In step 1 stakeholders were asked to identify the need for technology upgrade and prioritize the following factors when identifying the need for technology upgrades:

- Socio-economic factors;
- Capacity (Insufficient existing technologies);
- Environmental factors: GHG Reduction;

- Environmental factors: SO_x Reduction;
- Adaptation measures;
- Profitability;
- Increased efficiency; and
- Regulatory compliance.

In step 2 stakeholders were asked to select an appropriate technology for technology upgrade and prioritize the following factors when selecting the appropriate technology:

- Emission reduction factors;
- Return on Investment (ROI) (Cost Benefit Analysis);
- Availability of financial support;
- Ease of installation/ retrofit;
- Capacity (Insufficient existing technologies);
- Financial incentives;
- Upcoming regulations/ Future legal requirements; and
- Availability within the region.

4.4 Identification of Barriers to Technology Implementation

Possible barriers to the implementation of these technologies were identified and classified within initial research. Respondents were asked to rate the impact of the following barriers to the implementation of technology:

- Lack of Finance;
- Lack of Incentives;
- Technical Feasibility;
- No Guarantee of Return on Investment;
- No regulatory requirements;
- High Investment Cost;
- Organizational Challenges;
- Non-Availability of Technology in Region;
- Lack of Public-Private Partnerships;
- Financial Incentive; and
- Contractual Incentive.

Additionally, feedback received from the workshops related to technology was used in the analysis.

4.5 Identification of Measures to overcome these Barriers

Respondents were asked to list measures to overcome identified barriers, and all solutions were documented. Refer to appendices VI, VII, VIII, IX and X for the survey instruments disseminated to the targeted stakeholders via the use of the online survey platform “SurveyMonkey”.

5 Analysis of Results

MTCC Caribbean collected 30 responses from various stakeholders. Refer to appendices XI, XII, XIII and XIV for the individual results of each category of respondents. The analysis of the results are detailed below;

5.1 Previous Assessments

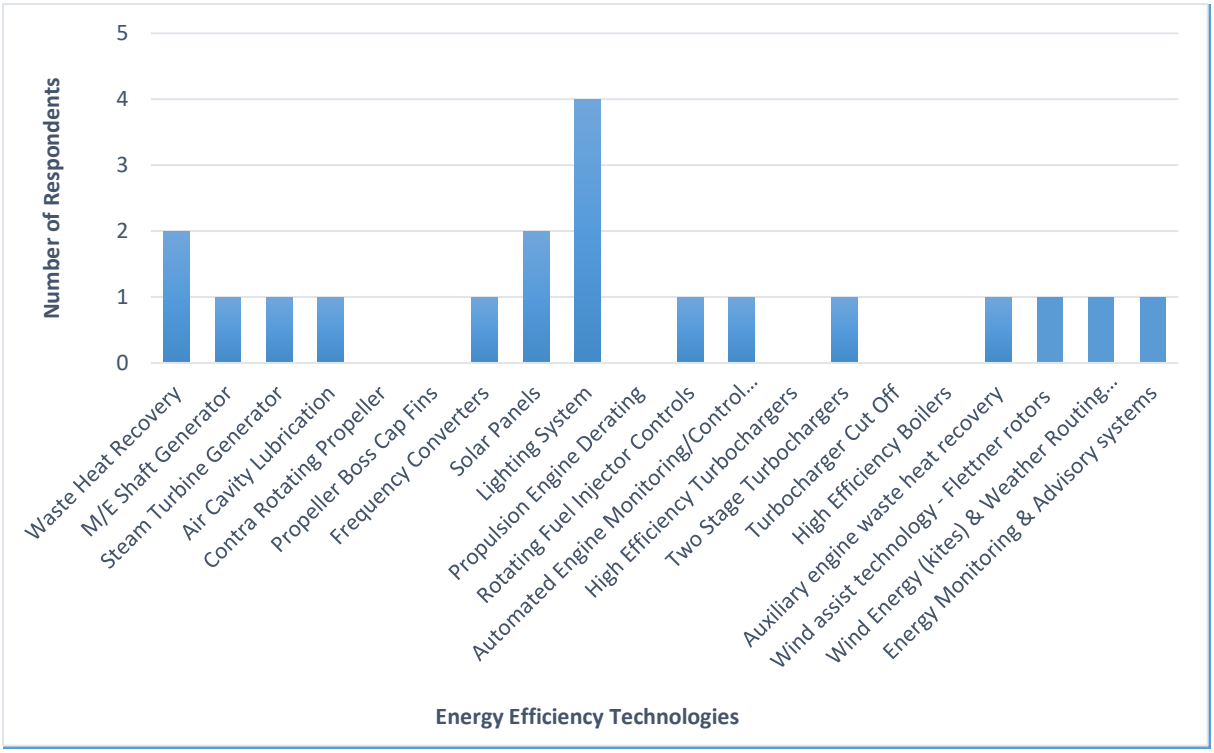
Jamaica and St. Kitts & Nevis Maritime Administrations indicated a TNA had been conducted within their country with respect to EE and emissions abatement methods. The other eight Maritime Administrations indicated there were no TNAs conducted within their country with respect to EE and emissions abatement methods.

5.2 Existing Technology

5.2.1 Energy Efficient Technology

Respondents identified the existing EE technologies available within the Caribbean for installation aboard vessels. Responses from the maritime administrations and the technology providers were used to determine the existing technologies as no responses were secured from shipyards. Figure 2 below provides a snapshot of the existing EE technology available for installation within the region.

Figure 2 - Existing Energy Efficient Technologies Available Within the Caribbean



Three maritime administrations and five technology providers, totaling eight respondents, identified 11 of the 17 technologies listed in the survey as available, in addition to 3 other technologies identified specifically by the technology providers which are as follows;

1. Wind assist technology - Flettner rotors
2. Wind Energy (kites) & Weather Routing System
3. Energy Monitoring & Advisory systems

The available EE technologies at shipyards within the Caribbean for installation aboard vessels amounts to 14 different types. Lighting systems, solar panels and waste heat recovery were the most commonly identified technologies. Notably, five of the eight maritime administrations that provided responses to this question clarified that there are no shipyards within their countries.

The ports and port authorities also indicated that there are no EE technologies at their facilities that are available for use by vessels that call at their ports.

5.2.2 Emission Abatement Technology

Respondents identified the existing emissions abatement technologies available within the Caribbean for installation aboard vessels. MTCC Caribbean listed the following six technologies for respondents to identify from;

- Selective Catalytic Reduction
- Exhaust Gas Scrubbers – Dry
- Exhaust Gas Scrubbers – Wet
- Humid Air Method
- Water Injection and Water Emulsion
- High Scavenge Pressure and Compression Ratio

Trinidad & Tobago identified Selective Catalytic Reduction as available within shipyards in their country while the other five maritime administrations that responded to the question indicated that there are no shipyards within their countries. The technology providers did not identify any emission abatement technologies available within the region. No responses were received from shipyards.

The ports and port authorities also indicated that there are no emission abatement technologies at their facilities that are available for use by vessels that call at their ports.

In contrast to energy efficiency technologies, there is limited availability of emission abatement technologies within the Caribbean region based on the responses from vessel owners/operators.

5.2.3 Presence of Energy Efficient/ Emission Abatement Technology Database

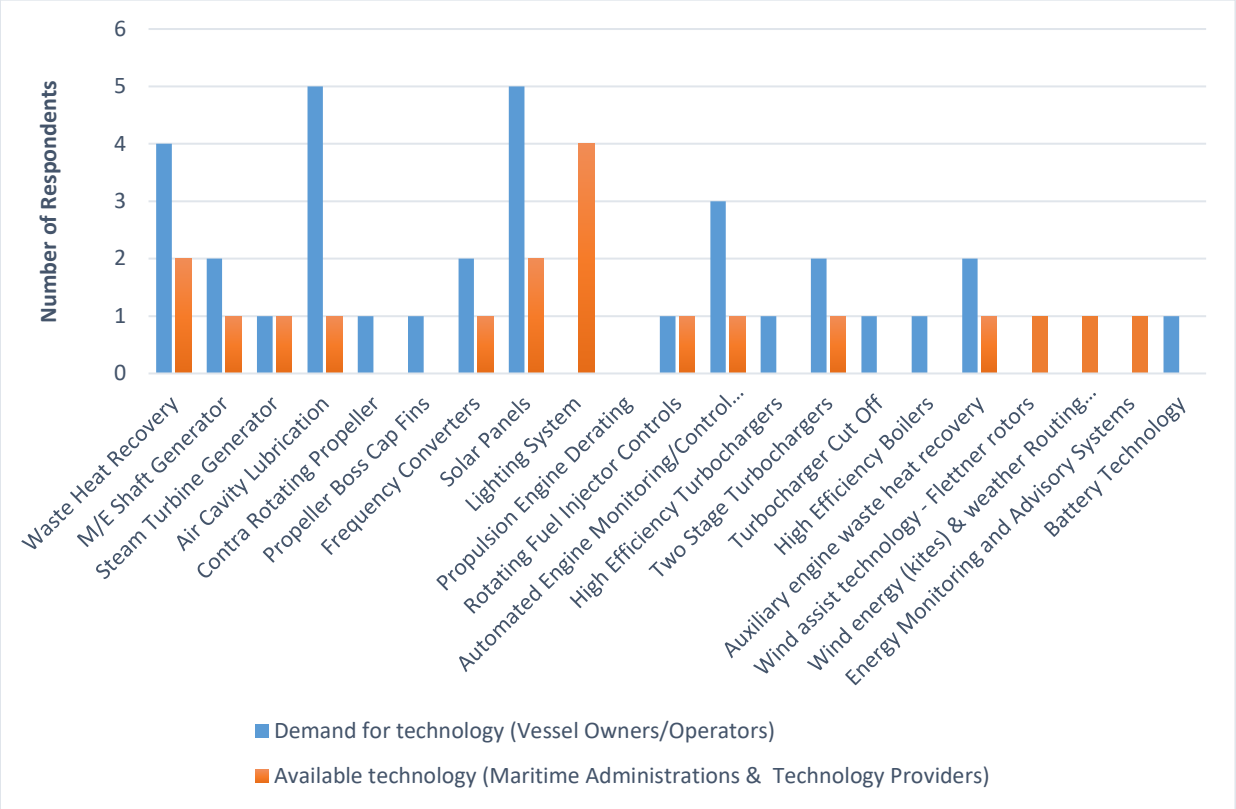
Belize Port Authority indicated that there is an existing database of Energy Efficient/ Emission Abatement technology on board vessels flagged to their country.

5.3 Demand for Technology

5.3.1 Demand for Energy Efficient Technology

The owners and operators identified EE technologies that are not available within shipyards in the Caribbean region, but which they would like installed/retrofitted on board their vessels. Figure 3 below compares the demand for EE technology as identified by vessel owners/operators against the technology available for installation aboard vessels, as identified by the maritime administrations and the technology providers within the Caribbean Region.

Figure 3 - Demand and Availability Comparison for energy efficient technologies



Eight of the eleven owners/operators provided a response to this question and identified 15 of the 17 energy efficient technologies listed in the survey as not being available at Caribbean shipyards for installation on board their vessels. This is however contrary to what is stated to be available by the maritime administrations and technology providers; they indicated that 14 different technologies are available within the region for installation aboard vessels. This shows a gap in the level of awareness of the owners/operators concerning what is available for installation in the Caribbean.

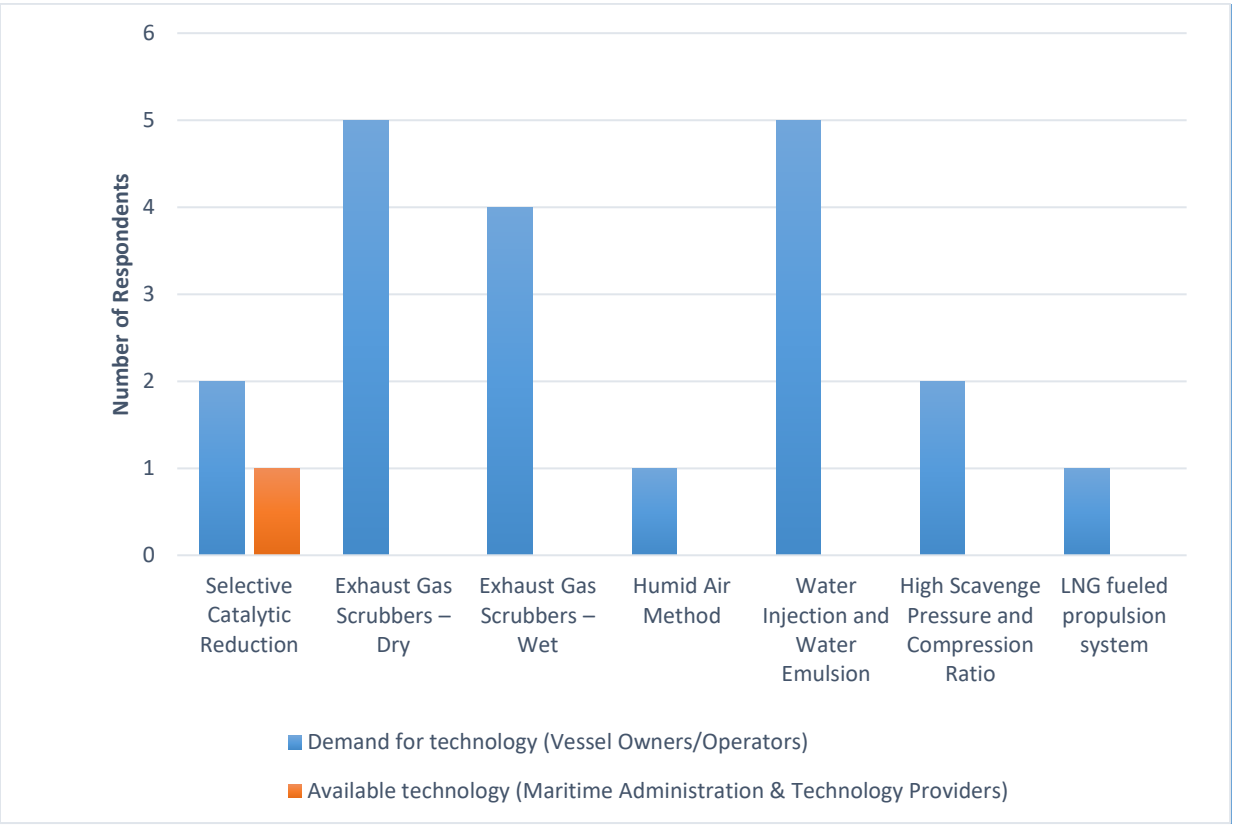
While there is neither a supply of nor demand for Propulsion Engine Derating, there is no demand by owners/operators for lighting systems, which is the most widely available technology at shipyards across the region.

While the maritime administrations and the technology providers identified solar panels and waste heat recovery among the more commonly available technologies, these technologies were identified among those most in demand by owners/operators that are currently unavailable at shipyards within the region. One vessel owner/operator also noted that they would like battery technology installed on their vessel but it is not available at shipyards within the region. One vessel owner/operator indicated that they have no demand for any EE technology available in the Caribbean region.

5.3.2 Demand for Emission Abatement Technology

The owners/operators identified emission abatement technologies that they would like to have for retrofitting their vessels but are currently unavailable in the Caribbean region. Figure 4 below compares the demand for emission abatement technology as identified by vessel owners/operators against the technology available for installation aboard vessels, as identified by the maritime administrations and the technology providers within the Caribbean Region.

Figure 4 - Demand and Availability Comparison for emission abatement technology



Eight of the eleven owners/operators provided a response to this question and identified all six of the emission abatement technologies listed by MTCC Caribbean as not being available at Caribbean shipyards for installation. This result is somewhat echoed by the one maritime administration that identified just one of the six technologies as being available in the region.

The emission abatement technologies in highest demand by owners and operators are;

1. Exhaust Gas Scrubbers – Dry

2. Exhaust Gas Scrubbers – Wet, and
3. Water Injection and Water Emulsion

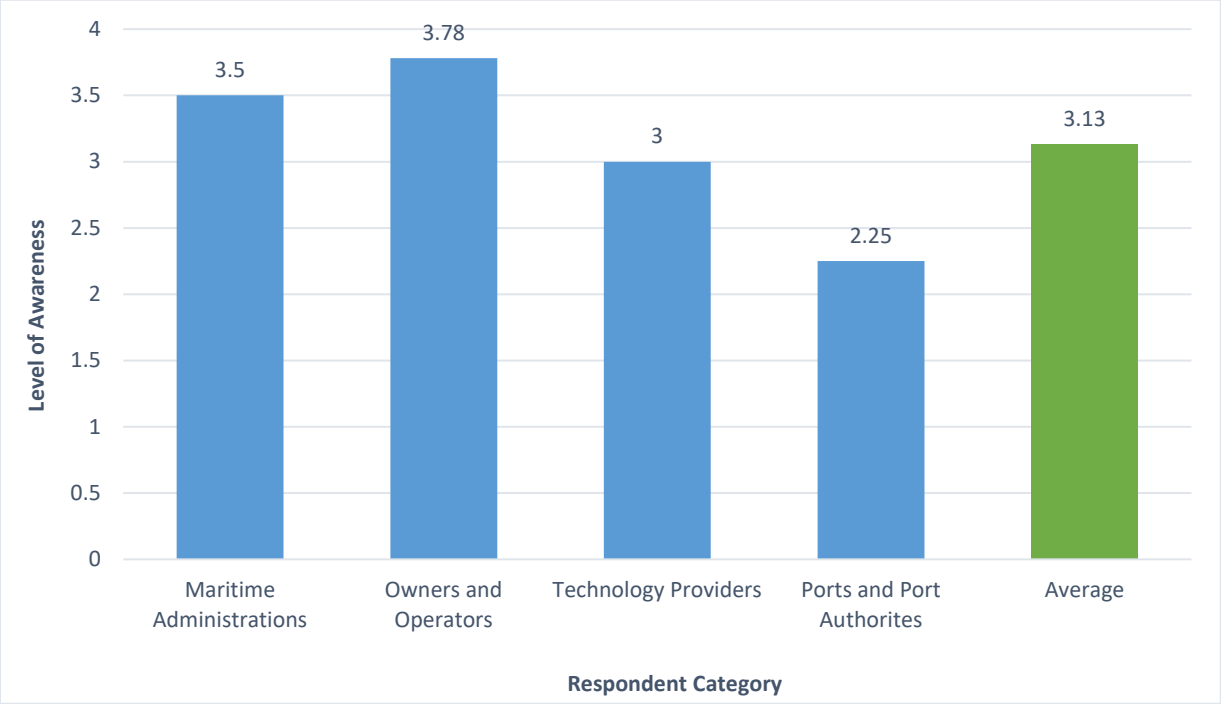
One owner/operator also identified an LNG fueled propulsion system separately as an emission abatement technology not available within the region that they would like installed on their vessel. One vessel owner/operator indicated that they have no demand for any emission abatement technology available in the Caribbean region.

Notably, the demand by owners/operators for emission abatement technologies is just as high as their demand for energy efficient technologies.

5.3.3 Awareness of Energy Efficient /Emission Abatement Technology

Demand for EE and emission abatement technology is affected by the level of awareness of those technologies. Respondents were therefore asked to indicate their level of awareness regarding the technologies using a scale of 1 to 5. The average awareness level of each category of respondent, totaling twenty-seven respondents, is illustrated in figure 5 below.

Figure 5 - Awareness of Energy Efficient /Emission Abatement Technology



Owners/operators are most aware of these technologies with a score of 3.78 out of 5, followed by maritime administration whom are responsible for regulation and enforcement surrounding the use of these technologies. The providers that design the technology are the third most aware category, followed by ports whose average awareness level is 2.25 out of 5; making them 60% less aware of the technologies than owners/operators. The ports also display the largest awareness gap among stakeholders.

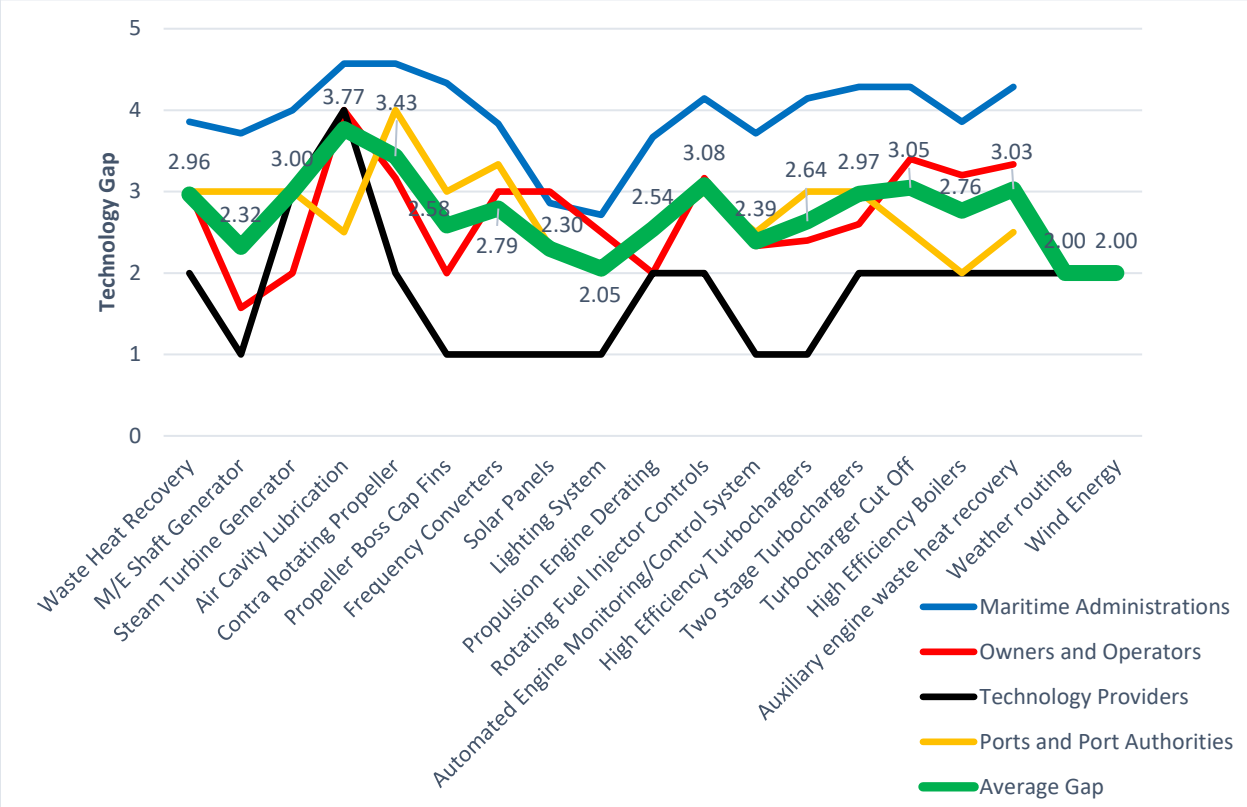
5.3.4 Gaps between technology available within the Caribbean and internationally available energy efficient technologies

The respondents identified the gaps between EE technology available within the Caribbean and energy efficient technologies available internationally using the following rating scale;

- 1 - very small gap
- 2 - small gap
- 3 - medium gap
- 4 - large gap
- 5 - very large gap

Eighteen respondents identified the gaps, and the average gaps for each category of respondents are illustrated in figure 6 below.

Figure 6 - Gap between technology available within the Caribbean and internationally available energy efficient technologies.



As shown in the figure above, the technology with the largest overall gap in availability is air cavity lubrication, while the technology with the lowest overall gap in availability is lighting systems; the latter of which was indicated to be highly available at shipyards within the Caribbean by the maritime administrations. While weather routing and wind technology show slightly lower gaps than lighting systems, these were not considered to be the most widely available as they were both identified by the one technology provider that provided a response to this question, while lighting systems were identified by all categories of respondents. The technology provider also

indicated the smallest gaps overall for available technology, while maritime administrations identified the largest gaps.

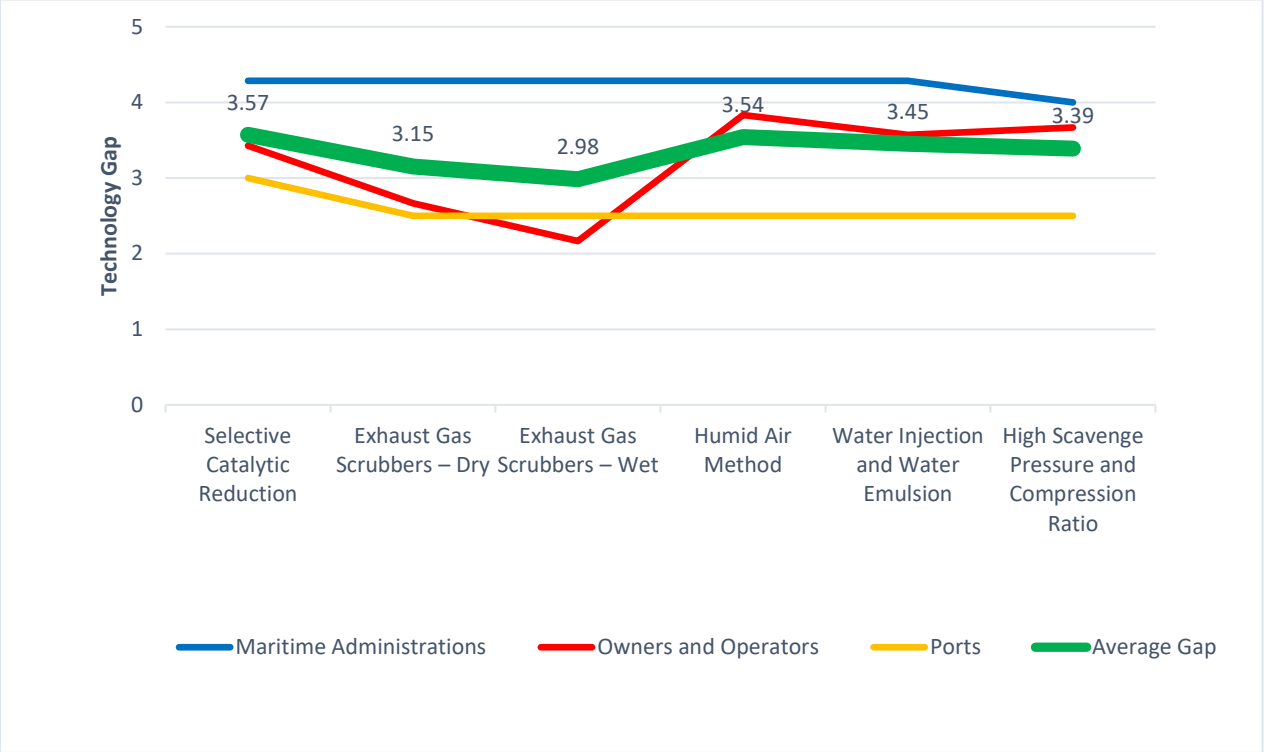
The gaps between the technology available within the Caribbean and internationally available technologies vary greatly across respondent categories, with very few similarities in availability across most technologies, showing great disparity among respondents concerning what is available. It should be noted that the owners/operators as well as ports show the smallest differences in their identification of technology gaps; in contrast, they showed the largest gap when identifying their awareness of EE and emission abatement technologies, with owners/operators being 60% more aware than the ports.

The overall average gap between technology available within the Caribbean and internationally available EE technologies is 2.80, which is closer to a medium gap.

5.3.5 Gaps between the Caribbean and internationally available emission abatement technologies

The respondents identified the gaps between emission abatement technology available in the Caribbean and emission abatement technologies available internationally using the previously outlined gap rating scale. Sixteen respondents identified the gaps, while no gaps were identified by the technology providers, as they had previously indicated that there are no emission abatement technologies available for retrofit/instalment on board vessels in the Caribbean region. The average gaps for each category of respondents are illustrated in figure 7 below.

Figure 7 - Gap between technology available within the Caribbean and internationally available emission abatement technologies.



The gaps identified among respondent categories show great disparity, with maritime administrations identifying the largest gaps, and ports identifying the smallest gaps. The largest technology gaps were identified for selective catalytic reduction and humid air method while the smallest gap was identified for exhaust gas scrubbers-wet. Across all technologies there is at a minimum, a small gap between what is available in the Caribbean and what is available internationally.

The overall average gap between technology available within the Caribbean and internationally available emission abatement technologies is 3.35, which is closer to a medium gap (3). This represents a higher average gap than that of energy efficient technologies which have an overall average gap of 2.80, amounting to 16.4% greater availability for these technologies than for emission abatement technologies.

5.4 Technology Needs

5.4.1 Factors considered in the determination of technology upgrades

Stakeholders consider a number of factors when identifying the need for technological upgrades to vessels. Twenty-two respondents across the four categories identified and ranked several factors in order of priority that apply when identifying the need for technological upgrades. These needs are considered under different scenarios for each category of respondents as follows;

1. Maritime Administrations – Vessels registered to their flag
2. Owners/Operators – Vessels they own/operate
3. Technology Providers – Vessels retrofitted by their company
4. Ports and Port Authorities – Vessels operating within their organization's jurisdiction.

The rankings range from 1 to 8, with 1 being the highest priority. The factors and their rankings are outlined in table 3 on the following page.

Table 3 - Factors considered in the determination of technology upgrades

Factors	Maritime Administrations									Owners and Operators									Tech Providers		Ports		Priority Total
	GY	**BZ**	**SKN**	**A&B**	**JA**	**SVG**	**CU**	**SE**	**TT**														
Socio-Economic Factors	2	7		2	3	4	7	3	8	2		1	1	8		8			8	4	8	2	4.59
Capacity (Insufficient existing technologies)	1		3	2	7	6	4	2	7	6		1		5		2			5	8	1	1	3.81
Environmental Factors: GHG Reduction	2			8	5	2	2		5	7	1	1	2	4	1	5	2		4	2	7		3.53
Environmental Factors: SOx Reduction	2			8	5	1	3	5	4	8	2	1	3	3	1	4	3		7	3	5		3.78
Adaptation Measures	2		4	2	6	5	5		6	5		1		7		1			6	7	6	5	4.53
Profitability	1	7	2	2	2		8		2	3	3	1		2		2	1	1	2	1	2		2.47
Increased Efficiency	1	8		5	4	7	6		3	4	3	1		6	1	3	4	2	3	5	3	3	3.79
Regulatory Compliance / Requirement	3	8	1	8	1	3	1	1	1	1	1	1		1	1	1		3	1	6	4	4	2.55

Of importance in the above results is the unusual rankings provided by six of the twenty-two respondents, where several numbers denoting the same ranking were used for different factors. These rankings affect the overall result of the ranking for the region, as the weighted average ranking of each factor is used to determine its priority for the region, with the lowest sum denoting the highest priority, and so forth. These irregularities are noted in the analysis, to provide some clarity.

Table 3 above illustrates the level of importance assigned to each factor among the four categories of respondents. Five of the maritime administrations that provided a priority ranking indicated that Regulatory Compliance/Requirement is of greatest importance when determining whether an upgrade is necessary. Three of these administrations are part of four administrations in total whom indicated that profitability is the second most important priority. Two of the administrations that provided irregular rankings appeared to use ranking number “8” to denote factors of high priority, rather than as a ranking for the factor of lowest priority. If this is taken to be the case, then seven of the administrations indicated that Regulatory Compliance/Requirement is the factor of greatest priority when determining whether a technology upgrade is necessary.

The rankings for the owners/operators also echoed the responses of the administrations, with Regulatory Compliance/Requirement and profitability being the factors of 1st and 2nd priority respectively. One owner/operator ranked all factors as a top priority, which may indicate that the respondent views all factors as equally important. There was no consensus among the technology providers concerning the most important factor, while one of the technology providers that did not provide a ranking clarified that they don't fit vessels for equipment, but their services are driven by cost savings and mandatory compliance, which is in line with the trends shown by the other respondents. The ports identified Capacity (Insufficient existing technologies) as their factor of highest priority.

Using the weighted average method and noting the above-mentioned exceptions, the factors affecting stakeholders across the region when identifying the need for technological upgrades in order of priority are:

1. Profitability
2. Regulatory Compliance
3. Environmental Factors: GHG Reduction
4. Environmental Factors: SOx Reduction
5. Increased Efficiency
6. Capacity (Insufficient existing technologies)
7. Adaptation Measures
8. Socio-Economic Factors

5.4.2 Factors considered in the determination of technology upgrades

Stakeholders consider a number of factors when selecting the appropriate technology for vessels after determining that an upgrade is necessary. Twenty-one respondents across the four categories identified and ranked several factors in order of priority that apply when selecting appropriate technology, taking into consideration the previously mentioned scenarios and priority ranking scale. The factors and their rankings are outlined in table 4 on the following page.

Table 4 - Factors considered in when selecting technology to perform upgrades

Factors	Maritime Administrations								Owners and Operators									Tech Providers		Ports		Priority Total
	GY	**BZ**	**A&B**	**JA**	**SVG**	**CU**	**SE**	**TT**														
Emission Reduction Factors	5		*	8	3	7	5		5	1	1			1	5	3		6	4		5	4.21
Return on Investment (ROI) (Cost Benefit Analysis)	1	8	*		4	8		7	2	1	1				1	2	1	1	1		2	2.86
Availability of Financial Support	1	7	*		5	1		6	4		1	1	3	1	1		3	2	2		1	2.60
Ease of Installation /Retrofit on existing vessels	2	7	*	5	2	5	3	4	3	2	1		4		1	1		5	5	4		3.38
Capacity (Insufficient existing technologies)	2		*	4	7	3	2	5	6		1				1			8	6	1	3	3.77
Financial Incentives	1	8	*		6	6		2	7	3	1	2			1		2	3	7		6	3.93
Upcoming Regulations/Future Legal Requirements	3	8	*	8	1	2	1	1	1	1	1	3	1	1	1	4	4	4	3	2	4	2.70
Availability within the Region	4		*		8	4	4	3	6	1	1		2		2	5		7	8	3	7	4.33

As with the previous analysis of priority rankings, eight of the twenty-one respondents ranked several factors using the same priority ranking. These will be discussed in the analysis.

There was less consensus among maritime administrations regarding the selection of technology when compared to determining whether an upgrade is necessary. Three of the eight maritime administrations ranked increased efficiency as their top priority, with that number increasing to five if the irregular ranking of “8” as the highest priority is factored in.

One of the maritime administrations, whose column is marked by a series of asterisks, highlighted all factors as being pertinent but offered no ranking, explaining that the ranking is more of a technical body question as to priority order. They went on to state that each of the factors could have equal weighting depending on the view point taken and the regulatory compliance mentality of the person assessing solutions and certifying approval manufacturers.

Return on Investment (ROI) (Cost Benefit Analysis) and Capacity (Insufficient existing technologies) scored equally using the weighted average method, ranking both as top priorities for owners and operators; though ROI could be considered the more important factor as it was identified by seven of the owners/operators, versus three for capacity. Notably, six of the owners/operators ranked Upcoming Regulations/Future Legal Requirements as their top priority, but the other three owners/operators ranked the factor as a less important priority. As previously done, one of the owners/operators ranked all factors as a top priority, which may indicate that the respondent views all factors as equally important.

The technology providers both indicated that Return on Investment (ROI) (Cost Benefit Analysis) and Availability of Financial Support are the most important factors to consider, while the ports identified Availability of Financial Support and Capacity (Insufficient existing technologies) as the most important factors. The technology provider that previously noted that their company does not fit vessels for equipment, provided rankings for the factors that affect the selection of appropriate technology; stating that the identified barriers and their impacts were selected in the context of providing services to shipping companies or national agencies.

Using the weighted average method and noting the above-mentioned exceptions, the factors affecting stakeholders across the region when selecting the appropriate technology for vessels in order of priority are:

1. Availability of Financial Support
2. Upcoming Regulations/Future Legal Requirements
3. Return on Investment (ROI) (Cost Benefit Analysis)
4. Ease of Installation /Retrofit on existing vessels
5. Capacity (Insufficient existing technologies)
6. Financial Incentives
7. Emission Reduction Factors
8. Availability within the Region

When the top priorities for stakeholders within the region are considered, both for deciding whether upgrades are necessary and when choosing the technology, the most important factors considered are financial viability and regulatory compliance.

5.5 Barriers

There are a number of barriers affecting stakeholders that impact the implementation of maritime technology transfer. The majority of these barriers are shared, with some specific to each category of respondent in line with their varying roles. The respondents rated the level of impact that each barrier had on their ability to implement maritime technology transfer using the following rating scale:

- 0 – No impact
- 1 – Low Impact
- 2 – Moderate Impact
- 3 – Medium Impact
- 4 – Significant Impact
- 5 – Major Impact

Twenty-three (23) respondents identified the barriers affecting maritime technology transfer. The barriers and their impact ratings are outlined in table 5 below.

Table 5 - Barriers impacting the implementation of maritime technology transfer

Barriers	*Maritime Administrations*	*Owners & Operators*	*Technology Providers*	*Ports*	*Overall*	*Comments from workshops*
Lack of Finance	Significant	Medium	Major	Major	Significant	Cuba, Barbados, Suriname, Grenada and St. Kitts & Nevis listed limited finances
Lack of Incentives	Medium	Moderate	Medium	Significant	Medium	Belize, Jamaica, Trinidad and St. Kitts & Nevis listed lack of incentives for ship-owners
Technology Feasibility	Significant	Medium	Medium	Medium	Medium	
No Guarantee on Return on Investment	Medium	Medium	Significant	Medium	Medium	
No Regulatory Requirements	Major	Significant	Significant	Medium	Significant	Jamaica listed national legislation and non-mandatory nature of fuel consumption and data requirements of SEEMP Trinidad & Tobago listed no legal requirements and St. Lucia listed lack of national legislation
High Investment Cost	Major	Medium	Major	Major	Major	Jamaica listed cost of hardware & system, and Trinidad & Tobago and St. Vincent and the Grenadines listed capital cost to ship-owners

Barriers	*Maritime Administrations*	*Owners & Operators*	*Technology Providers*	*Ports*	*Overall*	*Comments from workshops*
Organizational Challenges	Significant	Moderate	Moderate	Medium	Medium	Belize listed lack of equipment and technical expertise as a challenge Cuba indicated that technological training is required Haiti indicated the lack of political will to develop the maritime sector Barbados listed human resources and no technological facilities Guyana listed untrained professionals and lack of equipment Suriname listed lack of trained personnel St. Lucia listed lack of training and equipment
Non-Availability in the Region	Significant	Medium	Low	Medium	Medium	The Bahamas listed lack of energy efficient technology Trinidad & Tobago listed the availability of infrastructure
Lack of Public Private Partnership	Medium	Moderate	Moderate	Significant	Medium	
Financial Incentive	Not Applicable	Medium	Not Applicable	Not Applicable	Medium	
Contractual Incentive	Not Applicable	Medium	Not Applicable	Not Applicable	Medium	
Class Approvals	Not Applicable	Not Applicable	Moderate	Not Applicable	Moderate	

Barriers	*Maritime Administrations*	*Owners & Operators*	*Technology Providers*	*Ports*	*Overall*	*Comments from workshops*
Protection	Not Applicable	Not Applicable	Moderate	Not Applicable	Moderate	
Access to trials on ships	Not Applicable	Not Applicable	Medium	Not Applicable	Medium	

Based on the responses, *lack of finance* was identified as a major barrier by technology providers and ports. The category was rated as a significant barrier toward implementing maritime technology transfer. This result was also complemented by feedback from MTCC Caribbean's workshops where representatives from Cuba, Barbados, Suriname, Grenada and St. Kitts & Nevis all indicated limited availability of finance as a barrier. *Lack of incentives* was identified as a significant barrier by Ports/ Port Authorities and received a medium rating overall. Additionally, from the workshops, Belize, St. Kitts & Nevis, Jamaica and Trinidad & Tobago each identified lack of incentives to Ship owners/operators as being a barrier. *Technological feasibility* was identified as a significant barrier by the Maritime Administrations and received a medium rating overall. Feedback of note from the technology providers indicated that a significant barrier exists as a result of *no guarantee on the return on investment*.

Maritime Administrations identified *no regulatory requirements* as being a major barrier towards the implementation of maritime technology transfer. This result echoes the feedback from the workshops as Jamaica identified national legislation and non-mandatory nature of fuel consumption and data reporting requirements as a major challenge within their territory. St. Lucia identified lack of national legislation, and Trinidad & Tobago identified lack of legal framework as a barrier towards the implementation of maritime technology transfer. The only category to be rated as a major barrier overall was the *high investment cost* associated with maritime technology transfer. Maritime administrations, technology providers and ports each identified this factor as being a major barrier which was mirrored by the feedback received during workshops where Jamaica, St. Vincent and the Grenadines and Trinidad & Tobago each indicated that there is a high capital cost for ship owners.

Both *organizational challenges* and the *non-availability of technology within the region* were rated as medium barriers overall with only maritime administrations identifying this barrier as significant. Feedback coming from the workshops saw Belize, Cuba, Haiti, Barbados, Guyana, Suriname and St. Lucia each identify organizational challenges as a barrier to maritime technology transfer*. Lack of Public Private Partnership, Financial incentive* and *Contractual incentive* were each rated overall as a medium barrier. This was followed by *class approvals, protection,* and *access to trials on ships,* which were given ratings of moderate, moderate and medium respectively.

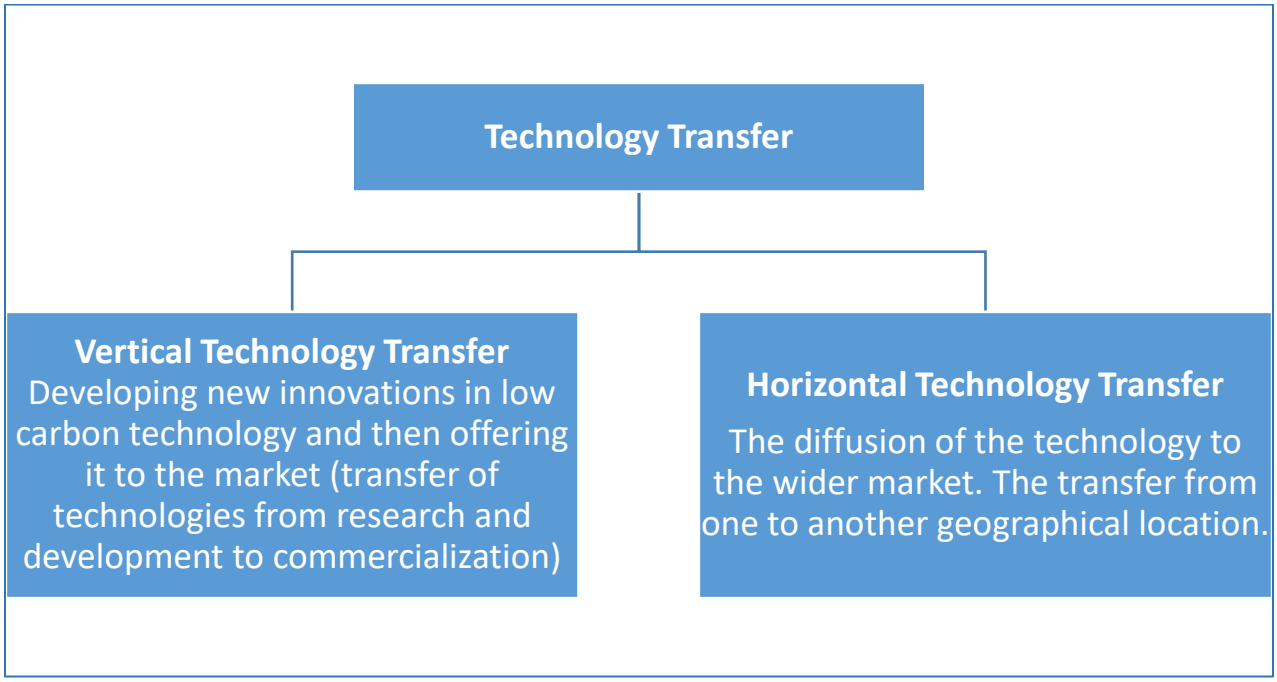
The feedback from this question identified the targeted barriers that will need to be addressed in the region to reduce the challenges associated with implementing maritime technology transfer.

6 Enabling Environment

Maritime technology developers are not based in the Caribbean. The majority of technology transfer is facilitated on a case by case basis between ship owners/operators and the technology providers. Although technology transfer is often a private interaction between two companies, institutional barriers and policies influence the transaction process, as well as the efficiency of the transfer process. Under perfect market conditions, all additional needs for energy services would be provided by the lowest cost measures, whether energy supply increases or energy demand reduces. There is considerable evidence that substantial energy efficiency investments are not made even when they are lower in cost than marginal energy supply. This condition therefore suggest that market barriers exist and there is therefore a need for policy intervention to create an enabling environment.

The enabling environment for technology transfer in the maritime sector involves a complex interface of human factor, institutional knowledge, institutional capacity and ships (type and age). In the case of low carbon technologies, especially maritime technologies, there are elements of both vertical and horizontal transfer, as illustrated in the figure 8 below, especially given the need for commercialization within the regional context. It is also imperative to state here that commercial acceptance within the maritime sector is influenced by logical business sense and mandatory requirements and compliance within the operating region. For technology transfer to be effective, the targeted receivers must be able to adopt, implement and sustain the technology. Adoption, implementation and sustainability of technology are linked to each other and the absence of any single component can lead to the failure of the maritime technology transfer process. The uncertainties of the maritime sector demand a more supporting environment when compared to other sectors.

Figure 8 - Avenues for technology transfer



To create a win-win scenario for technology adapters and providers, the creation of a genuine collaborative environment is required. Creating an enabling environment at each stage of the

value chain must be considered (Singh and Rambarath-Parasram, 2018) (Rambarath-Parasram, et al., 2018). It is not only important to lift the existing constraints but also to fill the gaps which exist in existing regulatory and administrative mechanisms. This situation may be resolved given the various initiatives specific to the maritime sector that have been brought by the IMO. In MARPOL Annex VI Regulation 23, emphasis is laid on co-operation along with transfer by promoting technical cooperation towards efficiency of ships. The enabling environment requires the following four components:

- Relevant Policy;
- Relevant Statutes;
- Institutional Responsibility for developing rules and implementing programmes, (which in some instances may involve developing existing and new institutional capacities), and
- Funding Mechanisms for capacity building and technology uptake.

Any successful policy reform supporting an enabling environment requires changes in implementation that facilitates compliance for targeted stakeholders. Such intervention will need to be cognizant of the incentives that drive business decisions. The development of the maritime sector is on the legislative agenda of many Caribbean territories. However, current top down approaches have failed to achieve desired results in the absence of support from the bottom level. The enabling environment therefore, should address the disparity between the available technologies and the technology in use, and will facilitate co-operation among regional administrations. Such cooperation should be directed toward the removal of barriers and adopt a framework which is efficient for receiving the required technology. This will require investment into institutions, infrastructure and the socio-economic environment. Capacity building is required to extract and absorb the complete benefits of the new technology adapted. Institutional capacities will need to be prepared accordingly.

While financial considerations and regulatory compliance are noted drivers for energy efficiency, the inadequacy of the enforcement of international standards in the region creates an environment where ship operators do not perceive the need to comply. Government intervention is likely to play a key role in facilitating low carbon technology transfer by developing a suitable policy. The business case for the uptake of energy efficiency can only be made when operating costs are reduced or capital costs reduced when compared to existing operational conditions. Most existing policies and programmes for the transfer of environmentally sound and greenhouse gas abatement technologies are addressed to national needs and rarely address the needs of the maritime sector, as it is not adequately included in many Caribbean NDCs.

Therefore to overcome the aforementioned challenges, future research is required to address the following:

- Complexity of the sector and address the interlinked nature of such factors (human and institutional);
- Absorption and Adaption of technology including investment into capacity building;
- Policy influence and institutional drivers, as major components of technology transfer;
- Organizational and Market barriers which affect the process of technology transfer; and
- Policy and Intellectual Property Rights (IPR) regime.

7 Technology Action Plan

The Technology Action Plan (TAP) is a plan for the uptake and diffusion of technologies based on identified priorities that will contribute to maritime climate change mitigation and adaptation. Therefore, this TAP is made up of numerous specific actions. The TAP is not technology-specific, but covers a portfolio of technologies as assessed under the Technology Needs Assessment (TNA). The same set of Actions will benefit all technologies.

The Technology Action Plan is the final outcome of the TNA. This plan outlines pathways for implementation of technology transfer based on the feedback received in the TNA. The TAP captures the:

- Identification and description of required actions for implementing Technology Transfer
- Identification of activities needed to implement identified actions
- Stakeholder engagement for these activities

The stakeholders within the Caribbean region were engaged during the TNA to highlight various necessary actions to overcome the barriers to maritime technology transfer. The summary of the discussion is captured in figure 9 below.

Figure 9 – Actions to overcome barriers to maritime technology transfer



7.1 Overview of Barriers, Measures to overcome, Identification, and Description of specific Activities to support Actions

Table 6 below presents the measures to overcome barriers identified during the TNA. The activities are proposed to address the measures to overcome the identified barriers. The stakeholders, in addition to MTCC Caribbean, are identified for the implementation of the activity engagement.

Table 6 – Measures to Overcome Barriers

Category	Identified Barrier	Measure to Overcome Barrier	Activity	Stakeholders
Finance and Economic	Lack of Finance	Access to Finance	- Bring together finance providers and Technology adopters & developers - Undertake TBD activities in response to the identification of implementation gaps - Organize a workshop for technology financers	Ship Owners, Ship Operators, Technology Providers, Technology Financers, Shipyards.
	Lack of Incentives	Creation of incentives at the regional and national levels (Ship registration, Ports)	- Mapping of the incentives by Ports and administrations to promote EE technology - Organize for business to business (B2B) networking events with specific emphasis on incentives appealing to ship operators	Maritime Administrations, Ports, Ship Owners and Operators
	No Guarantee on Return on Investment	Create financial framework, providing security to ship owners	- Identify financial / commercial resources - Identify and organize technology providers with flexible payment options	Ship Owners, Ship Operators, Technology Providers, Technology Financers, Ship yards.

Category	Identified Barrier	Measure to Overcome Barrier	Activity	Stakeholders
	High Investment Cost	Financial framework with reduced CAPEX	- Identify and attract technology providers with flexible financial solutions - Organized B2B workshops with technology financers	Ship Owners, Ship Operators, Technology Providers, Technology Financers, Ship yards.
	Contractual Incentive	Incentives for EE measures in charter party contracts	- Organize B2B workshop and networking promoting incentives for EE ships in charter party contracts	Ship owners / operators, Ship charterers
Regulations	Inadequate policy, legal and regulatory framework	Maritime regulatory compliance and enforcement Up to date National Maritime Legislation Improve policy and enabling environment Strengthen regulatory framework	- Highlight the regulatory concern at the high level Senior Maritime Advisors regional meeting - Build capacity towards alternative means of adopting legislation aimed at EE in the maritime sector - Other identified policy and enabling environment activities	Maritime Administrations, Ship owners / operators, Ports, NGO's, civil society
Organizational	HR Capacity	Capacity building initiatives at National and Regional level. Expand capacity building initiatives and collaboration (focused on technology) Creation of Platform for sharing of information on EE measures among Administrations.	- Capacity building technical workshops. - Provide platform for collaboration among Maritime administrations	Maritime Administrations, Ports, Ship Owners and Operators, Technology Provider, Shipyards in the region

Category	Identified Barrier	Measure to Overcome Barrier	Activity	Stakeholders
	Lack of Awareness	Capacity Building and Collaboration	- Identify and promote Economic, Environmental, Social and Financial benefits	Maritime Administrations, Ports, Ship Owners and Operators, Technology Provider, Shipyards in the region
Technology	Technology Feasibility	Modification of technology to meet individual needs	- Arrange for B2B meetings using MTCC network - Compile and distribute existing ship-specific pre-feasibility data	Technology Provider, Ship owner / operator,
	Professional networks for technology exist but are insufficiently active in the region	Enhance Networking for the Technology providers, adapters and facilitators	- Provide platform for Technology Providers and Financers at the capacity building initiatives	Ship owner / operator, Technology Provider, Shipyards in the region
	Non-Availability in the Region	Set up of regional units Connect the Shipyards, Ship owners in the region to Technology providers	- Arrange for B2B meetings using MTCC network	Technology Provider, Shipyards in the region
Others	Access to trials on ships	Create a platform in the form of an alliance where Ship owners and Operators can provide trial opportunities for class approved technologies.	- Arrange for B2B meetings using MTCC network	Ship owner / operator, Technology provider

7.2 Planning and Management

The activities and stakeholder engagement for each measure will require the necessary resources, including costs and the funding needs for the specific Actions and Activities. Considering budget constraints, further consultation with stakeholders within the region is necessary. These activities can be grouped into two categories encompassing a Regional Approach in the Diffusion of EE Technologies and the Implementation of Specific Prioritized Technology.

7.2.1 A Regional Approach in the Diffusion of EE Technologies

A Regional Approach with the establishment of a programme covering all interrelated actions for the diffusion of EE technologies. These activities will require large scale funding and depend on international donors or public sector grants. These activities can be covered under capacity building initiatives at regional levels.

7.2.2 Implementation of Specific Prioritized Technology

The implementation of specific prioritized technology is expected to be sponsored by the stakeholders benefitting from such activity. Activities and Small scale technology uptake pilot projects are expected to provide solutions to the issues arising from this TNA.

7.2.3 Next Phase

The activities in the next phase must be aligned with the top priorities identified by stakeholders in this TNA, namely:

- Economics of Maritime Operations after the uptake of EE Technology
- Availability of Financial Support for uptake of EE technology
- Upcoming Regulations/Future Legal Requirements

Based on the stakeholder feedback, the awareness of the need for interventions to facilitate reduced emissions and the GHG reduction potential of EE technologies seems to be sufficiently covered by various past capacity building activities. It may be useful to explore the feasibility of focussed capacity building in one regional shipyard in EE technology transfer to facilitate a one stop shop for end users. This can be achieved through a pilot project in the next phase of the project.

The findings from the Pilot projects and feedback from various capacity building activities of the first phase will further refine the TAP activities.

8 Limitations

MTCC Caribbean encountered several challenges during the undertaking of the technology needs assessment which are as follows;

1. Low response rate to the surveys – Of the seventy three (73) targeted stakeholders, only thirty (30) responses were received.

2. Incomplete survey submissions – Some respondents did not answer all questions. One respondent explained that he did not have sufficient/accurate information to complete the survey due to the lack of cooperation within different government agencies in his country (Antigua & Barbuda).

3. Misinterpretation of the questions – Particular sections of the survey may have been confusing for some of the respondents, and they may have misinterpreted the priority ranking questions, which affected the results.

4. Communication of survey – The survey was conducted online, and as such a survey link was forwarded to participants to complete. There were instances where the email addresses used were no longer operational or email accounts were too full and links were not delivered. Additionally, challenges were also encountered concerning attempts to follow up using newly obtained operational email addresses and phone calls to stakeholders to request responses.

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10 Appendices

10.1 Appendix I – Survey respondents: Maritime Administration

Maritime Administration	
Organization	Country
Antigua and Barbuda Department of Marine Services and Merchant Shipping (ADOMS)	Antigua and Barbuda
Belize Port Authority	Belize
Maritime Safety Directorate Cuban Maritime Administration	Cuba
Guyana Maritime Administration Department	Guyana
Haiti Maritime Services & Navigation	Haiti
Maritime Authority of Jamaica	Jamaica
Department of Maritime Affairs	St. Kitts & Nevis
Maritime Administration	St. Vincent & the Grenadines
Maritime Authority Suriname	Suriname
Maritime Services Division	Trinidad & Tobago

10.2 Appendix II – Survey respondents: Vessel Owners/Operators

Vessel Owners & Operators	
Organization	Country
British Petroleum Trinidad and Tobago (BPTT)	Trinidad & Tobago
Delta Logistics Limited	Trinidad & Tobago
Intercontinental Shipping Limited	Trinidad & Tobago
Caribbean Feeder Services	Panama
Vertraco Shipping	Netherlands
Trinidad Contractors Limited	Trinidad & Tobago
Wallem Shipmanagement Ltd	Hong Kong
OSM Ship Management	Singapore
Pacific Basin Shipping (HK) Ltd.	Hong Kong

10.3 Appendix III – Survey respondents: Letter from Shipyard

CARIBBEAN DOCKYARD AND ENGINEERING SERVICES LIMITED

CARIBBEAN DOCKYARD AND ENGINEERING SERVICES LIMITED
P.O Box 2874 TTPost Chaguaramas, Caribbean Dockyard Complex, Port Chaguaramas, Western Main Road, Trinidad, W.I
Tel: (868) 634-4127/28/29 Fax: (868) 634-4130/625-1215 Web Site: www.ttdockyard.com E-mail: info@ttdockyard.com

Date: 07DEC18
Attn: MTCC Caribbean

To whom it may concern;

We appreciate the consideration provided to Caribbean Dockyard and Engineering Services Limited (CDESL) for your survey "Shipyards and Dry-docks Technology Needs Analysis".

It would appear that this document / exercise by its context is geared more towards ship-owners/ship-builders and ship-operators. At the present time therefore we are of the view that same has no bearing or application in relation to our status as a Port facility largely performing operations in the nature of ship repair.

Our company does however continue at ways in which we may improve on our carbon footprint and expand our services offered to our clientele.

Sincerely,
CARIBBEAN DOCKYARD AND ENGINEERING SERVICES LIMITED

Rear Admiral Richard Kelshall
CHIEF EXECUTIVE OFFICER

10.4 Appendix IV - Survey respondents: Technology Providers

Technology Providers	
Organization	Country
International Windship Association	London
Fulcrum Maritime Systems, CLS Group	London
Circle	Italy
SkySails	London
ABB	London

10.5 Appendix V – Survey respondents: Ports

Ports	
Organization	Country
Freeport Harbor Company	The Bahamas
Kingtown Port	St. Vincent & the Grenadines
Port Authority of Trinidad and Tobago	Trinidad and Tobago
Stevedoring Services Limited	Bermuda

10.6 Appendix VI – Survey Instrument: Maritime Administration

Maritime Administration Technology needs assessment

Participant Information

1. What is your full name?

_____

2. Please state the name of your organization

_____

3. Please state your position within the organization

_____

4. Please select the country that your organization operates within

_____

A. Existing technology

1. Has a Technology Needs Assessment been conducted within your country with respect to Energy Efficiency and Emissions Abatement methods?

2. Which Energy Efficient/ Emission Abatement technology is available within shipyards in your country?

3. Is there an existing database of energy efficient/ emission abatement technology on board vessels flagged to your country?

B. Technology Gap

1. On a scale of 1-5, with 1 being least aware and 5 being very aware, please rate your awareness of Energy Efficient/ Emission Abatement technology

    a. 1, 2, 3, 4, 5

2. On a scale of 1 -5, with 1 being the smallest gap and 5 being the largest gap, please rate the difference between the existing technology within the Caribbean region and the available technology globally with respect to the following technologies

    a. Energy Saving Technologies
        i. Waste Heat Recovery
        ii. M/E Shaft Generator
        iii. Steam Turbine Generator

iv. Air Cavity Lubrication
v. Contra Rotating Propeller
vi. Propeller Boss Cap Fins
vii. Frequency Converters
viii. Solar Panels
ix. Lighting System
x. Propulsion Engine Derating
xi. Rotating Fuel Injector Controls
xii. Automated Engine Monitoring/Control System
xiii. High Efficiency Turbochargers
xiv. Two Stage Turbochargers
xv. Turbocharger Cut Off
xvi. High Efficiency Boilers
xvii. Auxiliary engine waste heat recovery
xviii. Other _____

b. Emission Abatement Technology
   i. Selective Catalytic Reduction
   ii. Exhaust Gas Scrubbers – Dry
   iii. Exhaust Gas Scrubbers – Wet
   iv. Humid Air Method
   v. Water Injection and Water Emulsion
   vi. High Scavenge Pressure and Compression Ratio
   vii. Other _____

C. Technology Needs

1. Choose the following factors that apply when identifying the need for technological upgrades to vessels registered to your flag. In the space provided next to the factors that you selected, rank the factors in order of priority.

   c. Socio-Economic Factors
   d. Capacity (Insufficient existing technologies)
   e. Environmental Factors: GHG Reduction
   f. Environmental Factors: SOx Reduction
   g. Adaptation Measures
   h. Profitability
   i. Increased Efficiency
   j. Regulatory Compliance / Requirement
   k. Other _____

2. Choose the following factors that apply when selecting the appropriate technology for vessels registered to your flag. In the space provided next to the factors that you selected, rank the factors in order of priority.

   a. Emission Reduction Factors
   b. Rate of Interest (ROI) (Cost Benefit Analysis)
   c. Availability of Financial Support

d. Ease of Installation /Retrofit on existing vessels
e. Capacity (Insufficient existing technologies)
f. Financial Incentives
g. Upcoming Regulations/Future Legal Requirements
h. Availability within the Region
i. Other _____

D. Barriers to the implementation of technology to meet technology needs

1. Using the following scale ranging between 0 and 5:

   a. 0 – No impact
   b. 1 – Low Impact
   c. 2 – Moderate Impact
   d. 3 – Medium Impact
   e. 4 – Significant Impact
   f. 5 – Major Impact

   Rate the barriers affecting your institution that impact the implementation of maritime technology transfer.

   a. Lack of Finance
   b. Lack of Incentives
   c. Technical Feasibility
   d. No Guarantee of Return on Investment
   e. No regulatory requirements
   f. High Investment Cost
   g. Organizational Challenges
   h. Non-Availability of Technology in Region
   i. Lack of Public-Private Partnerships
   j. Other _____

E. Solutions to the barriers

1. In your opinion, what are the necessary actions to overcome the barriers to maritime technology transfer?

   _____

10.7 Appendix VII – Survey Instrument: Vessel Owners/Operators

Ship Owner and Operator Technology needs assessment

Participant Information

5. What is your full name?

---

6. Please state the name of your company

---

7. Please state your position within the company

---

8. Please select the country that your company operates within

---

9. Please select the category that best describes your company

   a) Vessel Owner
   b) Vessel Operator
   c) Vessel Owner and Operator

A. Existing technology

1. Which Energy Efficient/ Emission Abatement technology is unavailable within shipyards in the Caribbean region, which you would like to be installed/retrofitted on board your vessel?

   a. Energy Saving Technologies
      i. Waste Heat Recovery
      ii. M/E Shaft Generator
      iii. Steam Turbine Generator
      iv. Air Cavity Lubrication
      v. Contra Rotating Propeller
      vi. Propeller Boss Cap Fins
      vii. Frequency Converters
      viii. Solar Panels
      ix. Lighting System
      x. Propulsion Engine Derating
      xi. Rotating Fuel Injector Controls
      xii. Automated Engine Monitoring/Control System
      xiii. High Efficiency Turbochargers
      xiv. Two Stage Turbochargers

xv. Turbocharger Cut Off
xvi. High Efficiency Boilers
xvii. Auxiliary engine waste heat recovery
xviii. Other _____

b. Emission Abatement Technology
   i. Selective Catalytic Reduction
   ii. Exhaust Gas Scrubbers – Dry
   iii. Exhaust Gas Scrubbers – Wet
   iv. Humid Air Method
   v. Water Injection and Water Emulsion
   vi. High Scavenge Pressure and Compression Ratio
   vii. Other _____

B. Technology Gap

1. On a scale of 1-5, with 1 being least aware and 5 being very aware, please rate your awareness of EE/EA technology

   a. 1, 2, 3, 4, 5

2. On a scale of 1 -5, with 1 being the smallest gap and 5 being the largest gap, please rate the difference between the existing technology the Caribbean region and the available technology globally with respect to the following technologies

   a. Energy Saving Technologies
      i. Waste Heat Recovery
      ii. M/E Shaft Generator
      iii. Steam Turbine Generator
      iv. Air Cavity Lubrication
      v. Contra Rotating Propeller
      vi. Propeller Boss Cap Fins
      vii. Frequency Converters
      viii. Solar Panels
      ix. Lighting System
      x. Propulsion Engine Derating
      xi. Rotating Fuel Injector Controls
      xii. Automated Engine Monitoring/Control System
      xiii. High Efficiency Turbochargers
      xiv. Two Stage Turbochargers
      xv. Turbocharger Cut Off
      xvi. High Efficiency Boilers
      i. Auxiliary engine waste heat recovery
      ii. Other _____

b. Emission Abatement Technology
    i. Selective Catalytic Reduction
    ii. Exhaust Gas Scrubbers – Dry
    iii. Exhaust Gas Scrubbers – Wet
    iv. Humid Air Method
    v. Water Injection and Water Emulsion
    vi. High Scavenge Pressure and Compression Ratio
    vii. Other _____

C. Technology Needs

1. Choose the following factors that apply when identifying the need for technological upgrades to vessels owned/operated by your company. In the space provided next to the factors that you selected, rank the factors in order of priority.

    a. Socio-Economic Factors
    b. Capacity (Insufficient existing technologies)
    c. Environmental Factors: GHG Reduction
    d. Environmental Factors: SOx Reduction
    e. Adaptation Measures
    f. Profitability
    g. Increased Efficiency
    h. Regulatory Compliance / Requirement
    i. Other _____

2. Choose the following factors that apply when selecting the appropriate technology for vessels owned/operated by your company. In the space provided next to the factors that you selected, rank the factors in order of priority.

    a. Emission Reduction Factors
    b. Rate of Interest (ROI) (Cost Benefit Analysis)
    c. Availability of Financial Support
    d. Ease of Installation /Retrofit on existing vessels
    e. Capacity (Insufficient existing technologies)
    f. Financial Incentives
    g. Upcoming Regulations/Future Legal Requirements
    h. Availability within the Region
    i. Other _____

D. Barriers to the implementation of technology to meet technology needs

1. Using the following scale ranging between 0 and 5:

    a. 0 – No impact
    b. 1 – Low Impact
    c. 2 – Moderate Impact
    d. 3 – Medium Impact

e. 4 – Significant Impact
f. 5 – Major Impact

Rate the barriers affecting your company that impact the implementation of maritime technology transfer.

a. Lack of Finance
b. Lack of Incentives
c. Technical Feasibility
d. No Guarantee of Return on Investment
e. No regulatory requirements
f. High Investment Cost
g. Organizational Challenges
h. Non-Availability of Technology in Region
i. Lack of Public-Private Partnerships
j. Financial Incentive
k. Contractual Incentive
l. Other _____

E. Solutions to the barriers

1. In your opinion, what are the necessary actions to overcome the barriers to maritime technology transfer?

_____

10.8 Appendix VIII – Survey Instrument: Shipyards/Dry-docks

Shipyards and Dry-docks Technology needs assessment

Participant Information

10. What is your full name?

_____

11. Please state the name of your company

_____

12. Please state your position within the company

_____

13. Please select the country that your company operates within

_____

A. Existing technology

1. Which Energy Efficient/ Emission Abatement technology is available within your facility for installation/ retrofit on board vessels?

c. Energy Saving Technologies
   - xix. Waste Heat Recovery
   - xx. M/E Shaft Generator
   - xxi. Steam Turbine Generator
   - xxii. Air Cavity Lubrication
   - xxiii. Contra Rotating Propeller
   - xxiv. Propeller Boss Cap Fins
   - xxv. Frequency Converters
   - xxvi. Solar Panels
   - xxvii. Lighting System
   - xxviii. Propulsion Engine Derating
   - xxix. Rotating Fuel Injector Controls
   - xxx. Automated Engine Monitoring/Control System
   - xxxi. High Efficiency Turbochargers
   - xxxii. Two Stage Turbochargers
   - xxxiii. Turbocharger Cut Off
   - xxxiv. High Efficiency Boilers
   - xxxv. Auxiliary engine waste heat recovery
   - xxxvi. Other _____

d. Emission Abatement Technology
   - viii. Selective Catalytic Reduction
   - ix. Exhaust Gas Scrubbers – Dry

x. Exhaust Gas Scrubbers – Wet
xi. Humid Air Method
xii. Water Injection and Water Emulsion
xiii. High Scavenge Pressure and Compression Ratio
xiv. Other _____

B. Technology Gap

1. On a scale of 1-5, with 1 being least aware and 5 being very aware, please rate your awareness of EE/EA technology

   a. 1, 2, 3, 4, 5

2. On a scale of 1 -5, with 1 being the smallest gap and 5 being the largest gap, please rate the difference between the existing technology within the Caribbean region and the available technology globally with respect to the following technologies

   a. Energy Saving Technologies
      i. Waste Heat Recovery
      ii. M/E Shaft Generator
      iii. Steam Turbine Generator
      iv. Air Cavity Lubrication
      v. Contra Rotating Propeller
      vi. Propeller Boss Cap Fins
      vii. Frequency Converters
      viii. Solar Panels
      ix. Lighting System
      x. Propulsion Engine Derating
      xi. Rotating Fuel Injector Controls
      xii. Automated Engine Monitoring/Control System
      xiii. High Efficiency Turbochargers
      xiv. Two Stage Turbochargers
      xv. Turbocharger Cut Off
      xvi. High Efficiency Boilers
      xvii. Auxiliary engine waste heat recovery
      xviii. Other _____

   b. Emission Abatement Technology
      i. Selective Catalytic Reduction
      ii. Exhaust Gas Scrubbers – Dry
      iii. Exhaust Gas Scrubbers – Wet
      iv. Humid Air Method
      v. Water Injection and Water Emulsion
      vi. High Scavenge Pressure and Compression Ratio

vii. Other _____

C. Technology Needs

1. Choose the following factors that apply when identifying the need for technological upgrades to vessels retrofitted by your company. In the space provided next to the factors that you selected, rank the factors in order of priority.

    a. Socio-Economic Factors
    b. Capacity (Insufficient existing technologies)
    c. Environmental Factors: GHG Reduction
    d. Environmental Factors: SOx Reduction
    e. Adaptation Measures
    f. Profitability
    g. Increased Efficiency
    h. Regulatory Compliance / Requirement
    i. Other _____

2. Choose the following factors that apply when selecting the appropriate technology for vessels retrofitted by your company. In the space provided next to the factors that you selected, rank the factors in order of priority.

    a. Emission Reduction Factors
    b. Rate of Interest (ROI) (Cost Benefit Analysis)
    c. Availability of Financial Support
    d. Ease of Installation /Retrofit on existing vessels
    e. Capacity (Insufficient existing technologies)
    f. Financial Incentives
    g. Upcoming Regulations/Future Legal Requirements
    h. Availability within the Region
    i. Other _____

D. Barriers to the implementation of technology to meet technology needs

1. Using the following scale ranging between 0 and 5:

    a. 0 – No impact
    b. 1 – Low Impact
    c. 2 – Moderate Impact
    d. 3 – Medium Impact
    e. 4 – Significant Impact
    f. 5 – Major Impact

    Rate the barriers affecting your company that impact the implementation of maritime technology transfer.

    a. Lack of Finance

b. Lack of Incentives
c. Technical Feasibility
d. No Guarantee of Return on Investment
e. No regulatory requirements
f. High Investment Cost
g. Organizational Challenges
h. Non-Availability of Technology in Region
i. Lack of Public-Private Partnerships
j. Other _____

E. Solutions to the barriers

1. In your opinion, what are the necessary actions to overcome the barriers to maritime technology transfer?

   _____

10.9 Appendix XI – Survey Instrument: Technology Providers

Technology Provider Technology needs assessment

Participant Information

14. What is your full name?

_____

15. Please state the name of your company

_____

16. Please state your position within the company

_____

17. Please select the country that your company operates within

_____

A. Existing technology

1. Which energy Efficient/ Emission Abatement technology is available within the Caribbean region for installation/ retrofit on board vessels?

e. Energy Saving Technologies

xxxvii. Waste Heat Recovery
xxxviii. M/E Shaft Generator
xxxix. Steam Turbine Generator
xl. Air Cavity Lubrication
xli. Contra Rotating Propeller
xlii. Propeller Boss Cap Fins
xliii. Frequency Converters
xliv. Solar Panels
xlv. Lighting System
xlvi. Propulsion Engine Derating
xlvii. Rotating Fuel Injector Controls
xlviii. Automated Engine Monitoring/Control System
xlix. High Efficiency Turbochargers
l. Two Stage Turbochargers
li. Turbocharger Cut Off
lii. High Efficiency Boilers
liii. Auxiliary engine waste heat recovery
liv. Other _____

f. Emission Abatement Technology

xv. Selective Catalytic Reduction
xvi. Exhaust Gas Scrubbers – Dry
xvii. Exhaust Gas Scrubbers – Wet
xviii. Humid Air Method
xix. Water Injection and Water Emulsion
xx. High Scavenge Pressure and Compression Ratio
xxi. Other _____

2. Where are these Energy Efficient/ Emission Abatement technologies available? (Please list the country/ shipyard/ drydock/ port)

B. Technology Gap

1. On a scale of 1-5, with 1 being least aware and 5 being very aware, please rate your awareness of EE/EA technology

   a. 1, 2, 3, 4, 5

2. On a scale of 1 -5, with 1 being the smallest gap and 5 being the largest gap, please rate the difference between the existing technology within the Caribbean region and the available technology globally with respect to the following technologies

   g. <u>Energy Saving Technologies</u>
      lv. Waste Heat Recovery
      lvi. M/E Shaft Generator
      lvii. Steam Turbine Generator
      lviii. Air Cavity Lubrication
      lix. Contra Rotating Propeller
      lx. Propeller Boss Cap Fins
      lxi. Frequency Converters
      lxii. Solar Panels
      lxiii. Lighting System
      lxiv. Propulsion Engine Derating
      lxv. Rotating Fuel Injector Controls
      lxvi. Automated Engine Monitoring/Control System
      lxvii. High Efficiency Turbochargers
      lxviii. Two Stage Turbochargers
      lxix. Turbocharger Cut Off
      lxx. High Efficiency Boilers
      lxxi. Auxiliary engine waste heat recovery
      lxxii. Other _____

   h. <u>Emission Abatement Technology</u>
      xxii. Selective Catalytic Reduction
      xxiii. Exhaust Gas Scrubbers – Dry
      xxiv. Exhaust Gas Scrubbers – Wet
      xxv. Humid Air Method
      xxvi. Water Injection and Water Emulsion
      xxvii. High Scavenge Pressure and Compression Ratio

xxviii. Other _____

C. Technology Needs

1. Choose the following factors that apply when identifying the need for technological upgrades to vessels retrofitted by your company. In the space provided next to the factors that you selected, rank the factors in order of priority.

    a. Socio-Economic Factors
    b. Capacity (Insufficient existing technologies)
    c. Environmental Factors: GHG Reduction
    d. Environmental Factors: SOx Reduction
    e. Adaptation Measures
    f. Profitability
    g. Increased Efficiency
    h. Regulatory Compliance / Requirement
    i. Other _____

2. Choose the following factors that apply when selecting the appropriate technology for vessels retrofitted by your company. In the space provided next to the factors that you selected, rank the factors in order of priority.

    a. Emission Reduction Factors
    b. Rate of Interest (ROI) (Cost Benefit Analysis)
    c. Availability of Financial Support
    d. Ease of Installation /Retrofit on existing vessels
    e. Capacity (Insufficient existing technologies)
    f. Financial Incentives
    g. Upcoming Regulations/Future Legal Requirements
    h. Availability within the Region
    i. Other _____

D. Barriers to the implementation of technology to meet technology needs

2. Using the following scale ranging between 0 and 5:

    a. 0 – No impact
    b. 1 – Low Impact
    c. 2 – Moderate Impact
    d. 3 – Medium Impact
    e. 4 – Significant Impact
    f. 5 – Major Impact

    Rate the barriers affecting your institution that impact the implementation of maritime technology transfer.

a. Lack of Finance
b. Lack of Incentives
c. Technical Feasibility
d. No Guarantee of Return on Investment
e. No regulatory requirements
f. High Investment Cost
g. Organizational Challenges
h. Non-Availability of Technology in Region
i. Lack of Public-Private Partnerships
j. Class Approvals
k. Protection
l. Access to Trials on Ships
m. Other _____

E. Solutions to the barriers

1. In your opinion, what are the necessary actions to overcome the barriers to maritime technology transfer?

   _____

10.10 Appendix X – Survey Instrument: Ports

Ports Technology needs assessment

Participant Information

18. What is your full name?

_____

19. Please state the name of your institution

_____

20. Please state your position within the institution

_____

21. Please select the country that your institution operates within

_____

F. Existing technology

1. Which Energy Efficient/ Emission Abatement technology is available within your facility for use by vessels calling at your port?

G. Technology Gap

1. On a scale of 1-5, with 1 being least aware and 5 being very aware, please rate your awareness of EE/EA technology

    a. 1, 2, 3, 4, 5

2. On a scale of 1 -5, with 1 being the smallest gap and 5 being the largest gap, please rate the difference between the existing technology the Caribbean region and the available technology globally with respect to the following technologies

    a. Energy Saving Technologies
        i. Waste Heat Recovery
        ii. M/E Shaft Generator
        iii. Steam Turbine Generator
        iv. Air Cavity Lubrication
        v. Contra Rotating Propeller
        vi. Propeller Boss Cap Fins
        vii. Frequency Converters
        viii. Solar Panels
        ix. Lighting System

x. Propulsion Engine Derating
xi. Rotating Fuel Injector Controls
xii. Automated Engine Monitoring/Control System
xiii. High Efficiency Turbochargers
xiv. Two Stage Turbochargers
xv. Turbocharger Cut Off
xvi. High Efficiency Boilers
xvii. Auxiliary engine waste heat recovery
xviii. Other _____

b. Emission Abatement Technology
   i. Selective Catalytic Reduction
   ii. Exhaust Gas Scrubbers – Dry
   iii. Exhaust Gas Scrubbers – Wet
   iv. Humid Air Method
   v. Water Injection and Water Emulsion
   vi. High Scavenge Pressure and Compression Ratio
   vii. Other _____

H. Technology Needs

1. Choose the following factors that apply when identifying the need for technological upgrades to vessels operating within the jurisdiction of your organization. In the space provided next to the factors that you selected, rank the factors in order of priority.

   a. Socio-Economic Factors
   b. Capacity (Insufficient existing technologies)
   c. Environmental Factors: GHG Reduction
   d. Environmental Factors: SOx Reduction
   e. Adaptation Measures
   f. Profitability
   g. Increased Efficiency
   h. Regulatory Compliance / Requirement
   i. Other _____

2. Choose the following factors that apply when selecting the appropriate technology for vessels operating within the jurisdiction of your organization. In the space provided next to the factors that you selected, rank the factors in order of priority.

   a. Emission Reduction Factors
   b. Rate of Interest (ROI) (Cost Benefit Analysis)
   c. Availability of Financial Support
   d. Ease of Installation /Retrofit on existing vessels
   e. Capacity (Insufficient existing technologies)
   f. Financial Incentives
   g. Upcoming Regulations/Future Legal Requirements
   h. Availability within the Region

i. Other _____

I. Barriers to the implementation of technology to meet technology needs

3. Using the following scale ranging between 0 and 5:

   a. 0 – No impact
   b. 1 – Low Impact
   c. 2 – Moderate Impact
   d. 3 – Medium Impact
   e. 4 – Significant Impact
   f. 5 – Major Impact

   Rate the barriers affecting your institution that impact the implementation of maritime technology transfer.

   n. Lack of Finance
   o. Lack of Incentives
   p. Technical Feasibility
   q. No Guarantee of Return on Investment
   r. No regulatory requirements
   s. High Investment Cost
   t. Organizational Challenges
   u. Non-Availability of Technology in Region
   v. Lack of Public-Private Partnerships
   w. Other _____

J. Solutions to the barriers

2. In your opinion, what are the necessary actions to overcome the barriers to maritime technology transfer?

_____

10.11 Appendix XI – Survey results: Maritime Administration

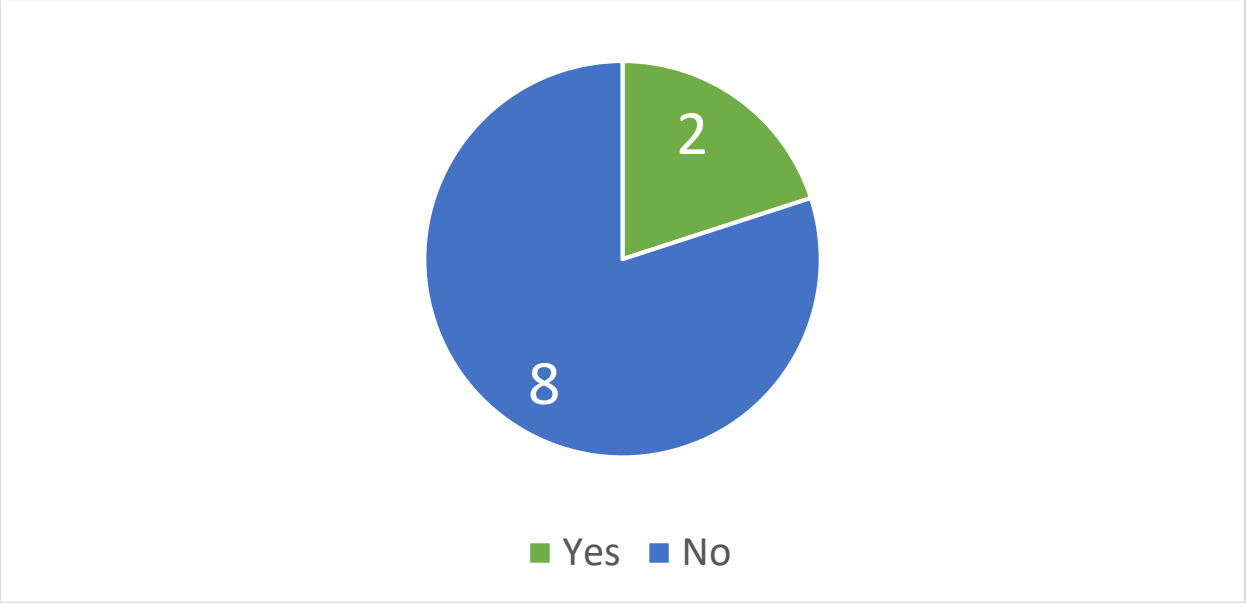
MTCC Caribbean received survey responses from ten (10) maritime administrations, which are detailed below.

10.11.1 Existing Technology

10.11.1.1 Past Technology Needs Assessments

The ten (10) respondents indicated whether a technology needs assessment has been conducted within their country with respect to Energy Efficiency and Emissions Abatement methods, which is illustrated in figure 10 below.

Figure 10 - Past Technology Needs Assessments

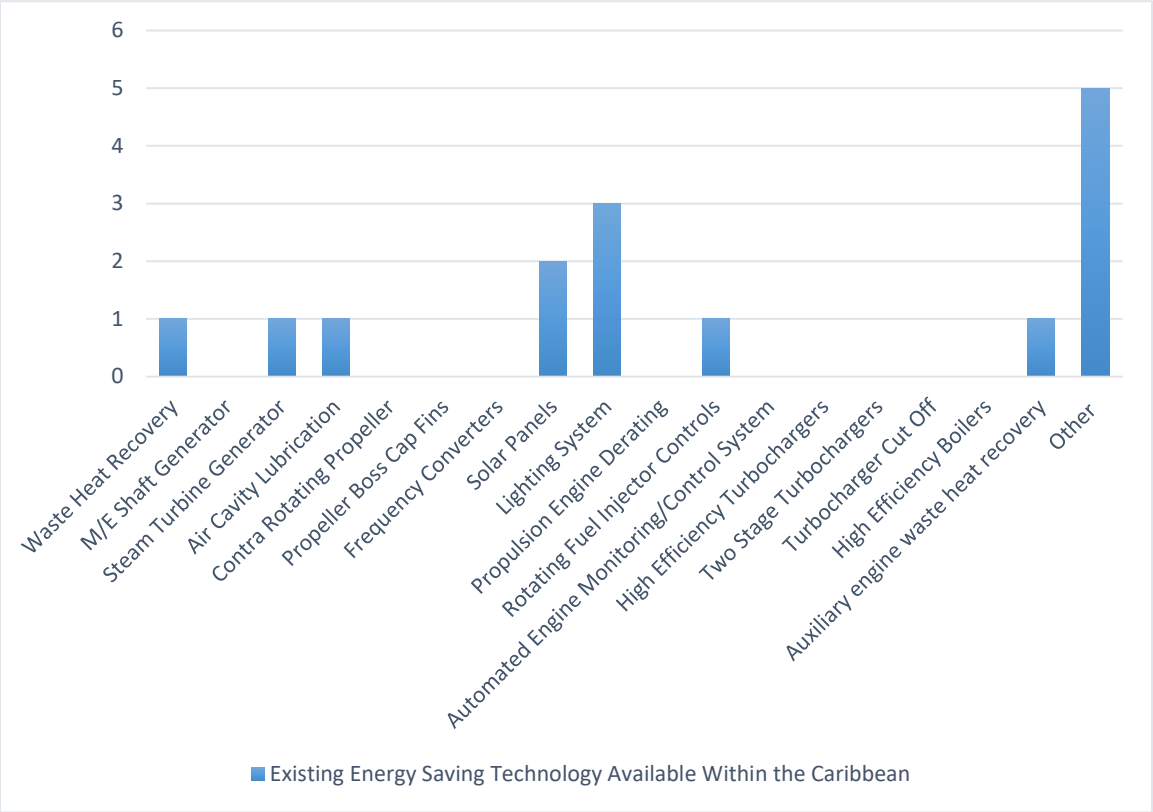


Eight (8) of the ten (10) respondents indicated that no technology needs assessment regarding Energy Efficiency and Emissions Abatement methods has been conducted within their country, while two (2) respondents indicated that an analysis has been conducted. (St. Kitts and Belize)

10.11.1.2 Energy Efficient Technology

Eight (8) of the ten (10) respondents identified energy efficient technologies available at shipyards within their countries for installation aboard vessels and two (2) respondents did not answer the question. The results are illustrated in figure 11 below.

Figure 11 - Existing Energy Efficient Technology Available Within the Caribbean



Of the 17 listed energy efficient technologies, the maritime administrations indicated that 7 are available at shipyards within their countries for installation aboard vessels.

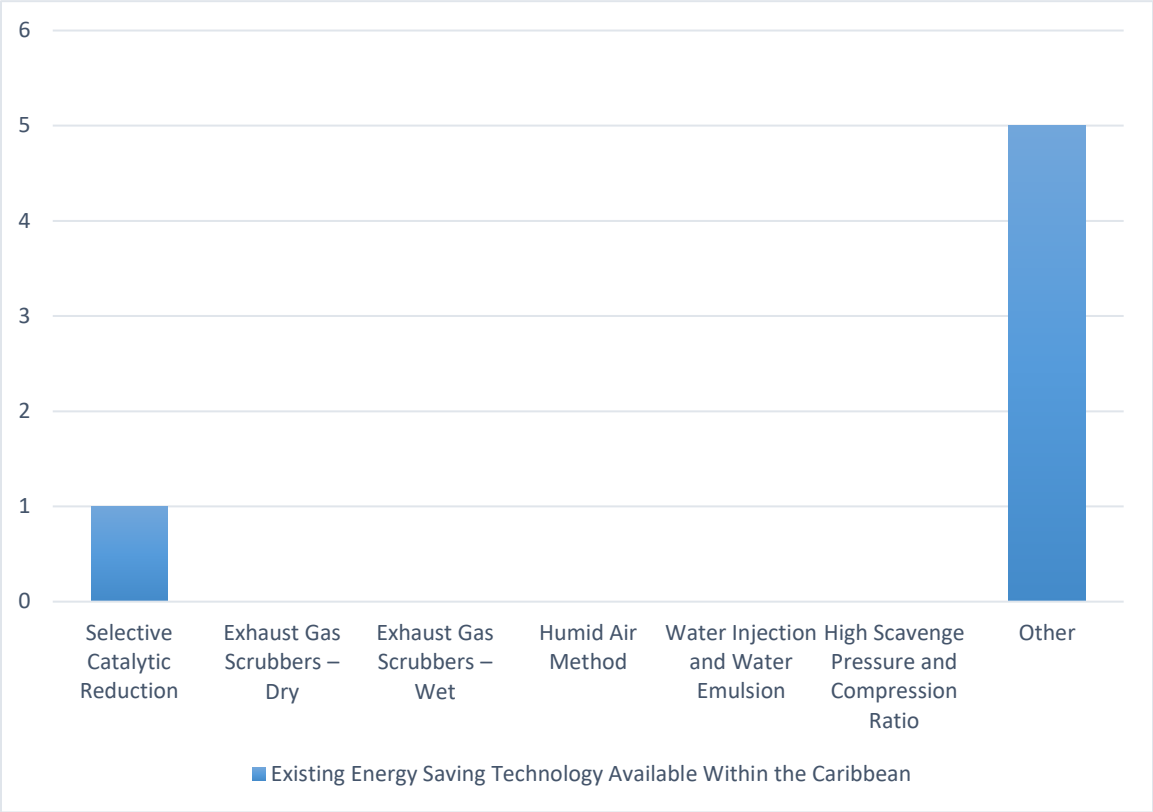
Five (5) of the eight (8) respondents chose the "other" option, and their responses were as follows;

1. None, we have no ship yards (St. Kitts)
2. No shipyards and not aware of any of above at local boatyard (Antigua)
3. We do not have shipyard in Haiti (Haiti)
4. No active shipyards in the country: The build locations produce artisanal craft while the repair facilities accommodate mainly fishing boats small tugs and small motor yachts. (Jamaica)
5. None (Suriname)

10.11.1.3 Emission Abatement Technology

Six (6) of the ten (10) respondents identified emission abatement technologies available at shipyards within their countries for installation aboard vessels and four (4) respondents did not answer the question. The results are illustrated in figure 12 below.

Figure 12 - Existing Emission Abatement Technology Available Within the Caribbean



Of the 6 listed emissions abatement technologies, one (1) of the maritime administrations indicated that 1 of the technologies is available at shipyards within their country for installation aboard vessels.

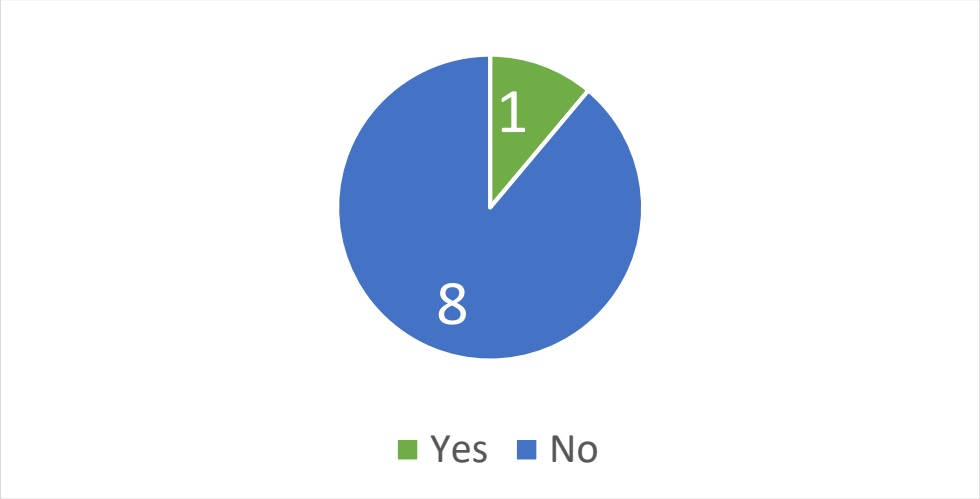
Five (5) of the six (6) respondents chose the “other” option, and their responses were as follows;

1. The ship yard in my country don't have any of the above technologies available is mainly for storage (St. Kitts)
2. No shipyards and not aware of any of above at local boatyard (Antigua)
3. None (Belize)
4. No active shipyards in the country: The build locations produce artisanal craft while the repair facilities accommodate mainly fishing boats small tugs and small motor yachts. (Jamaica)
5. None (Suriname)

10.11.1.4 Presence of Energy Efficient/ Emission Abatement Technology Database

Nine (9) of the ten (10) respondents indicated whether there is an existing database of Energy Efficient/ Emission Abatement technology on board vessels flagged to their country and one (1) respondent did not answer the question. The results are illustrated in figure 13 below.

Figure 13 - Presence of Energy Efficient/ Emission Abatement Technology Database



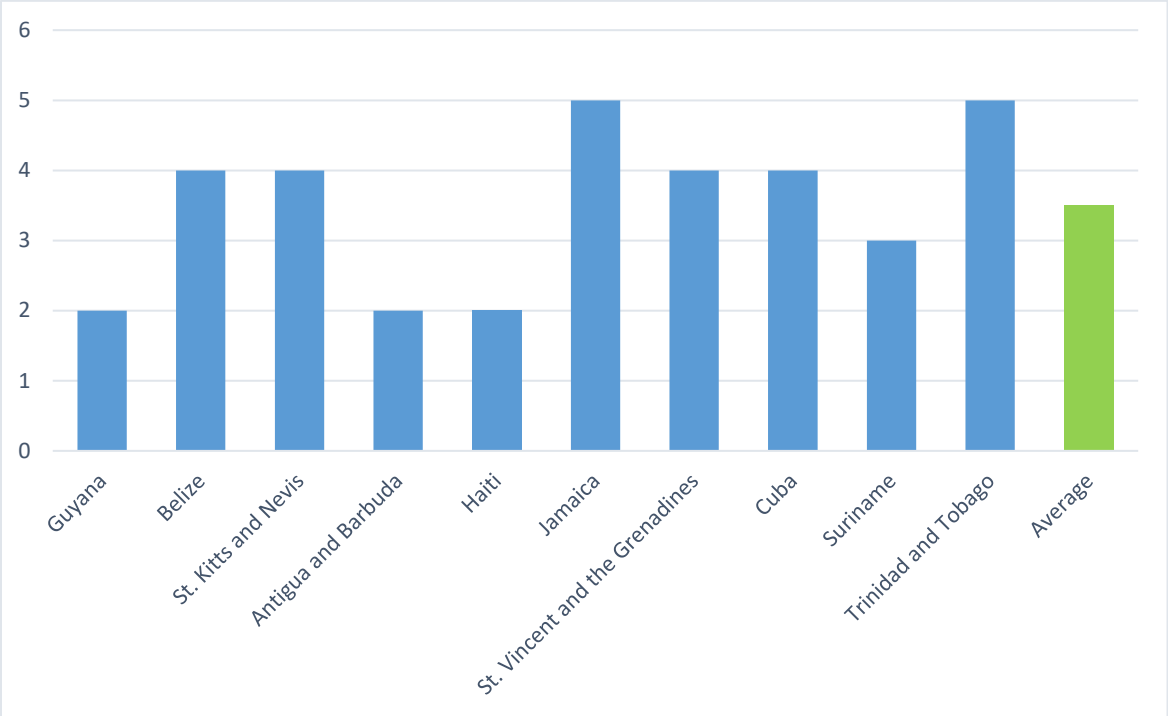
Eight (8) of the Nine (9) respondents indicated that a database of energy efficient/emission abatement technology on board vessels flagged to their country does not currently exist, while one (1) respondent (Belize) indicated that a database does in fact exist within their country.

10.11.2 Technology Gap

10.11.2.1 Awareness of Energy Efficient /Emission Abatement Technology

The ten (10) respondents indicated on a scale of 1 to 5, their level of awareness concerning Energy Efficient/Emission Abatement Technology, which is illustrated in figure 14 below.

Figure 14 - Awareness of Energy Efficient /Emission Abatement Technology



The respondents displayed varying levels of awareness with the average awareness for Maritime Administrations being 3.5 out of 5.

10.11.2.2 Gap between technology available within the Caribbean and internationally available energy efficient technologies

The respondents were asked to identify the gaps between energy efficient technology available within the Caribbean and energy efficient technologies available globally using the following rating scale;

- 1 - very small gap
- 2 - small gap
- 3 - medium gap
- 4 - large gap
- 5 - very large gap

Seven (7) of the ten (10) respondents identified the gaps, while three (3) respondents chose the "other" option. The technology gaps are illustrated in figure 15 below.

Figure 15 - Gap between technology available within the Caribbean and internationally available energy efficient technologies.



The gaps between energy efficient technologies available within the Caribbean and energy efficient technology available globally ranged from 1 to 5, varying widely among respondents. Notably, one respondent identified very large gaps for every category. The average gap for all technologies identified by the respondents is 3.93, which is closer to a large gap (4).

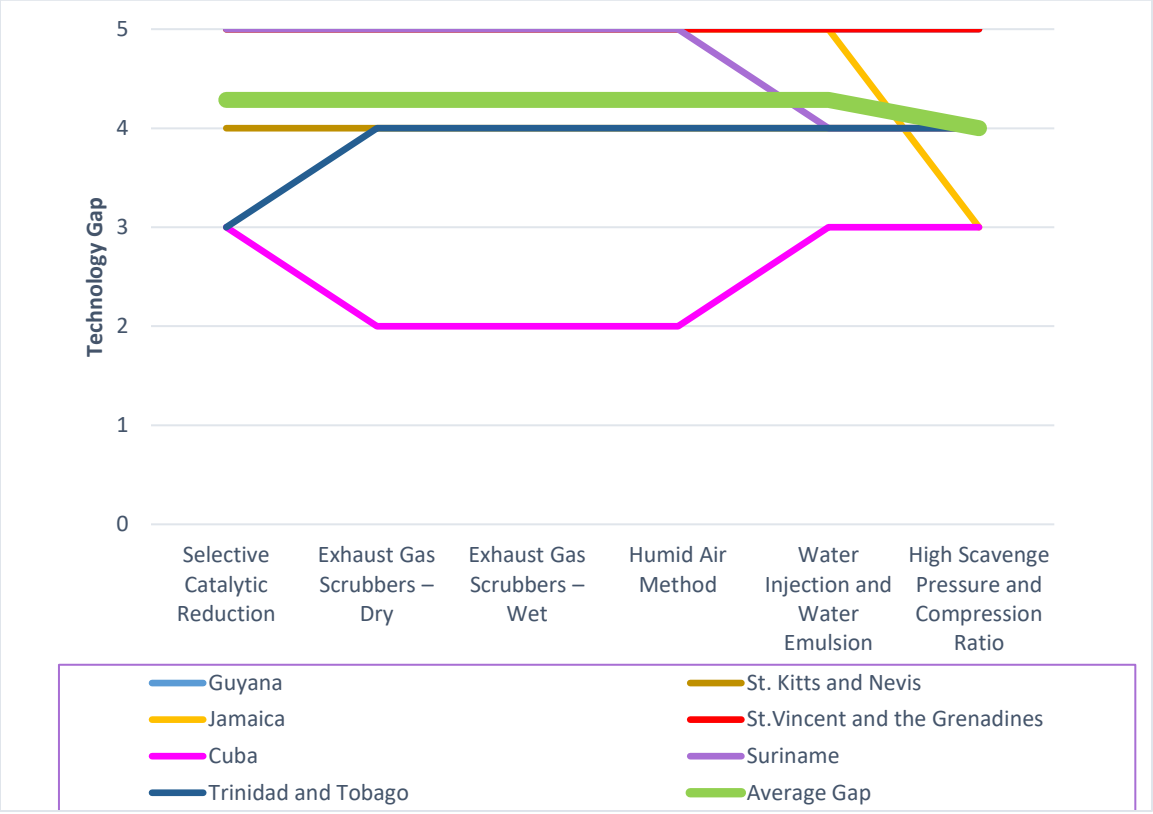
Three (3) of the respondents chose the “other” option, and their responses were as follows;

1. NA (Belize)
2. I have no access to information to answer this question (Antigua)
3. there is no such technologies existing in Haiti (Belize)

10.11.2.3 Gap between technology available within the Caribbean and internationally available emission abatement technology

The respondents were asked to identify the gaps between emission abatement technology available within the Caribbean and emission abatement technologies using the previously outlined gap rating scale. Seven (7) of the ten (10) respondents identified the gaps, while three (3) respondents chose the “other” option. The technology gaps are illustrated in figure 16 below.

Figure 16 - Gap between technology available within the Caribbean and internationally available emission abatement technologies.



The gaps between emission abatement technologies available within the Caribbean and emission abatement technology available globally ranged from 2 to 5, varying among respondents. The average gap for all technologies identified by the respondents is 4.23, which is closer to a large gap (4).

Three (3) of the respondents chose the “other” option, and their responses were as follows;

1. NA (Belize)
2. I have no access to information to answer this question (Antigua)
3. there is no such technologies existing in Haiti (Belize)

10.11.3 Technology Needs

10.11.3.1 Factors considered in the determination of technology upgrades

Nine (9) of the ten (10) respondents identified and ranked several factors in order of priority that apply when identifying the need for technological upgrades to vessels registered to their flag. One (1) respondent chose the “other” option. The factors are outlined in table 7 below.

Table 7 - Factors considered in the determination of technology upgrades

Factors	**Guyana**	**Belize**	**St. Kitts and Nevis**	**Antigua and Barbuda**	**Jamaica**	**St. Vincent and the Grenadines**	**Cuba**	**Suriname**	**Trinidad and Tobago**
Socio-Economic Factors	2	7		2	3	4	7	3	8
Capacity (Insufficient existing technologies)	1		3	2	7	6	4	2	7
Environmental Factors: GHG Reduction	2			8	5	2	2		5
Environmental Factors: SOx Reduction	2			8	5	1	3	5	4
Adaptation Measures	2		4	2	6	5	5		6
Profitability	1	7	2	2	2		8		2
Increased Efficiency	1	8		5	4	7	6		3
Regulatory Compliance / Requirement	3	8	1	8	1	3	1	1	1

Five (5) of the nine (9) respondents prioritized Regulatory Compliance / Requirement as their top priority, while three (3) of the nine (9) prioritized profitability as the second most important factor.

One (1) respondent chose the “other” option, and their response is as follows;

1. I think that Haiti being a coastal State, the Port State Control part should be considered not only the Flag State Control side. It is very important for us not being a flag state (Haiti)

10.11.3.2 Factors considered when selecting technology to perform upgrades

Eight (8) of the ten (10) respondents identified and ranked several factors in order of priority that apply when selecting the appropriate technology for vessels registered to their flag. One (1) respondent identified all of the factors but did not provide a ranking, choosing to clarify using the “other” option, and the last respondent did not answer the question. The factors are outlined in table 8 below.

Table 8 - Factors considered when selecting technology to perform upgrades

Factors	**Guyana**	**Belize**	**Antigua and Barbuda**	**Jamaica**	**St. Vincent and the Grenadines**	**Cuba**	**Suriname**	**Trinidad and Tobago**
Emission Reduction Factors	5		*	8	3	7	5	
Return on Investment (ROI) (Cost Benefit Analysis)	1	8	*		4	8		7
Availability of Financial Support	1	7	*		5	1		6
Ease of Installation /Retrofit on existing vessels	2	7	*	5	2	5	3	4
Capacity (Insufficient existing technologies)	2		*	4	7	3	2	5
Financial Incentives	1	8	*		6	6		2
Upcoming Regulations/Future Legal Requirements	3	8	*	8	1	2	1	1
Availability within the Region	4		*		8	4	4	3

Three (3) of the nine (9) respondents prioritized Upcoming Regulations/Future Legal Requirements as their top priority.

The respondent that identified all of the factors but did not provide a ranking (column marked with an asterisk) chose to clarify using the "other" option, and their response is as follows;

1. This is more of a technical body question as to priority order. Each of the factors could have equal weight depending on the view point taken and the regulatory compliance mentality of the person assessing solutions and certifying approval manufacturers.

10.11.4 Barriers to the implementation of technology to meet technology needs

10.11.4.1 Barriers impacting the implementation of maritime technology transfer

The respondents were asked to rate the barriers affecting their institution that impact the implementation of maritime technology transfer using the following rating scale;

- 0 – No impact
- 1 – Low Impact
- 2 – Moderate Impact
- 3 – Medium Impact
- 4 – Significant Impact
- 5 – Major Impact

The ten (10) respondents identified the barriers affecting maritime technology transfer. One respondent, having selected 2 of the 9 factors, clarified using the "other" option. The barriers and their impact ratings are outlined in table 9 below.

Table 9 - Barriers impacting the implementation of maritime technology transfer

Factors	**Guyana**	**Belize**	**St. Kitts and Nevis**	**Antigua and Barbuda**	**Haiti**	**Jamaica**	**St. Vincent and the Grenadines**	**Cuba**	**Suriname**	**Trinidad and Tobago**
Lack of Finance	5	4	5		4	2	5	5	4	4
Lack of Incentives	5	4	1		5	3		2	3	4
Technical Feasibility	5	3	4	1	5	3		4	4	3
No Guarantee of Return on Investment	5	5	1	0	2	3		2	3	3
No regulatory requirements	5	4	5		5	5	5	2	5	4

High Investment Cost	5	5	4		5	4		5	4	3
Organizational Challenges	5	4	4		5	1	5	4	1	2
Non-Availability of Technology in Region	5	3	4		4	2	5	4	3	2
Lack of Public-Private Partnerships	5	3	4		5	1		5	3	1

The respondent that identified 2 of the barriers chose to clarify using the "other" option, and their response is as follows;

1. I do not have access to information to fully answer this.

Another respondent chose the "other" option to add clarification to their impact rating, stating the following;

2. Again, we are not a Flag State but it is very important for Haiti to be trained and have technologies as we are a developing country and as 90% of good being used come to us from the maritime transport. Above all, 87% of taxes collected by the Ministry of Finance, are drawn from the maritime sector

10.11.5 Solutions to the barriers to maritime technology transfer

The respondents were asked what they believe are the necessary actions to overcome the barriers to maritime technology transfer and the following was stated by the ten (10) respondents.

1. Maritime regulatory compliance and enforcement, incentives, ease of uptake of technology, ease of retrofitting, greater availability of technology
2. Up to date National Maritime Legislation, Availability of Incentives, Cooperation between various stakeholders (Public-Private) and Cooperation between the Various Maritime Authorities
3. To Awareness the institutions and persons in charge of implement and to execute the actions to improve the Energy Efficiency and to mitigate the emissions of GHG. 2. To establish an effective cooperation among the countries of the region in relation to the exchange of knowledge and technologies to increase Energy Efficiency and to reduce the Emissions of SOx and NOx. 3. Developed countries cooperation with the countries in development and small States islanders for the financing technological needs and the preparedness of the qualified persons.
4. Implementation of the SVG Shipping Pollution Prevention Act and its Regulations. On-going discussions/negotiations with the Ministry of Finance and National Security for needed finances.
5. The regulatory framework and the tools to enforce the international standards; Strong flag legislation instrument or flag/class rules to hasten the pace of technology adoption and transfer; The availability of incentives and cost effective energy efficiency solutions.

6. The first on for me should be the raising of awareness on the importance of the maritime sector for a developing Island state by bringing knowledge on what the maritime sector is made of. For that, the MTCC could consider coming to Haiti to raise not only awareness on the maritime sector but also consider technical training related to the field.
7. Closer cooperation from other Government Agencies
8. Conducive learning process and common expectations regarding the technology in question. Supplementing this with expert and stakeholder interviews either directly or by using questionnaires and conducting a workshop with key stakeholders.
9. The necessary actions to overcome barriers of maritime technology transfer is more responsibility for the flag to offer incentives to their flagged vessels. Incentives at ports may not be feasible since all ports may not be able to offer incentives so the vessels may not be bothered to employ better technologies if only one or two ports visited provides some form of incentives. On the other hand, since the majority of the countries with registries have also signed on to various commitments to combat climate change, this would be a prime opportunity to show their commitment to the cause. Seeing as the industry is moved by profits and if there is no assured return of investment, technology transfer may have to be incentivized and the flags are in a strategic position to see this happen.
10. Affordability of the necessary technology, along with substantial incentives for implementing new technologies.

10.12 Appendix XII – Survey results: Vessel Owners and Operators

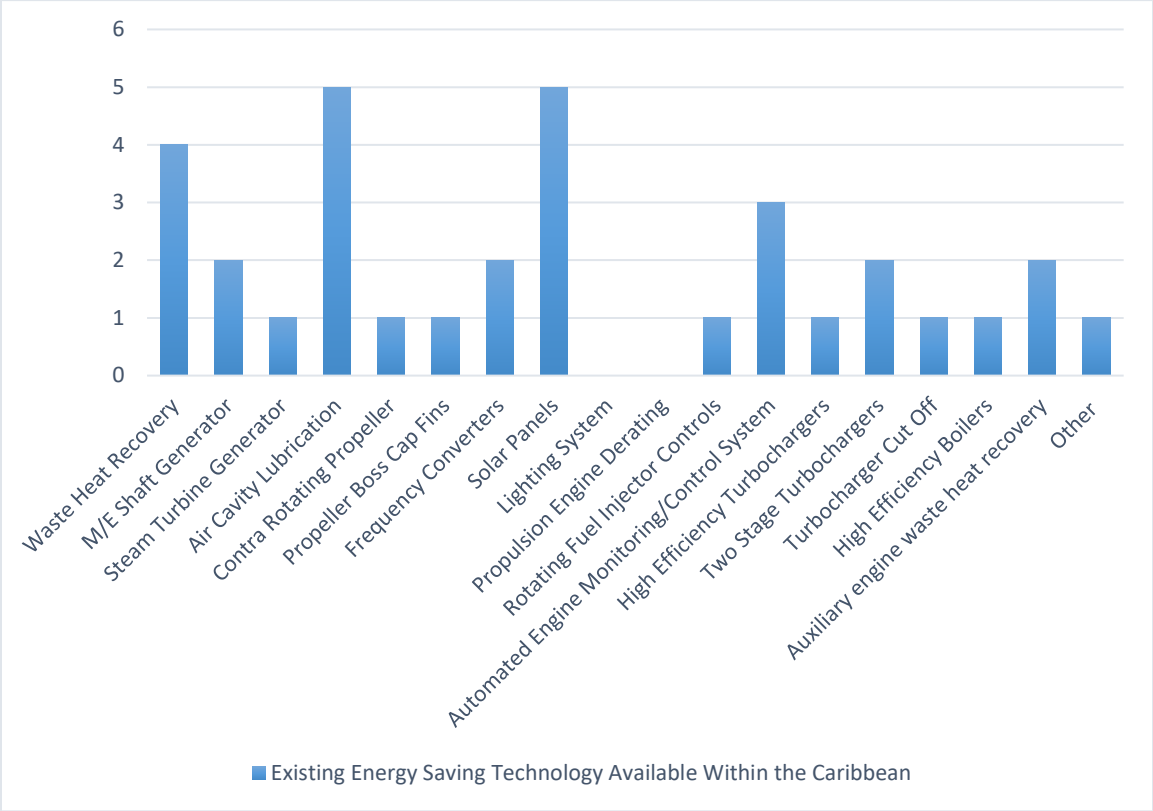
MTCC Caribbean received survey responses from eleven (11) Owners & Operators, which are detailed below.

10.12.1 Existing Technology

10.12.1.1 Energy Efficient Technology

Eight (8) of the eleven (11) respondents identified energy efficient technology that is not available within shipyards in the Caribbean region, which they would like installed/retrofitted on board their vessels. The technologies are illustrated in figure 17 below.

Figure 17 - Existing Energy Efficient Technology



Of the 17 listed energy efficient technologies, the owners and operators identified 15 technologies not available within shipyards in the Caribbean region that they would installed/retrofitted on board their vessels.

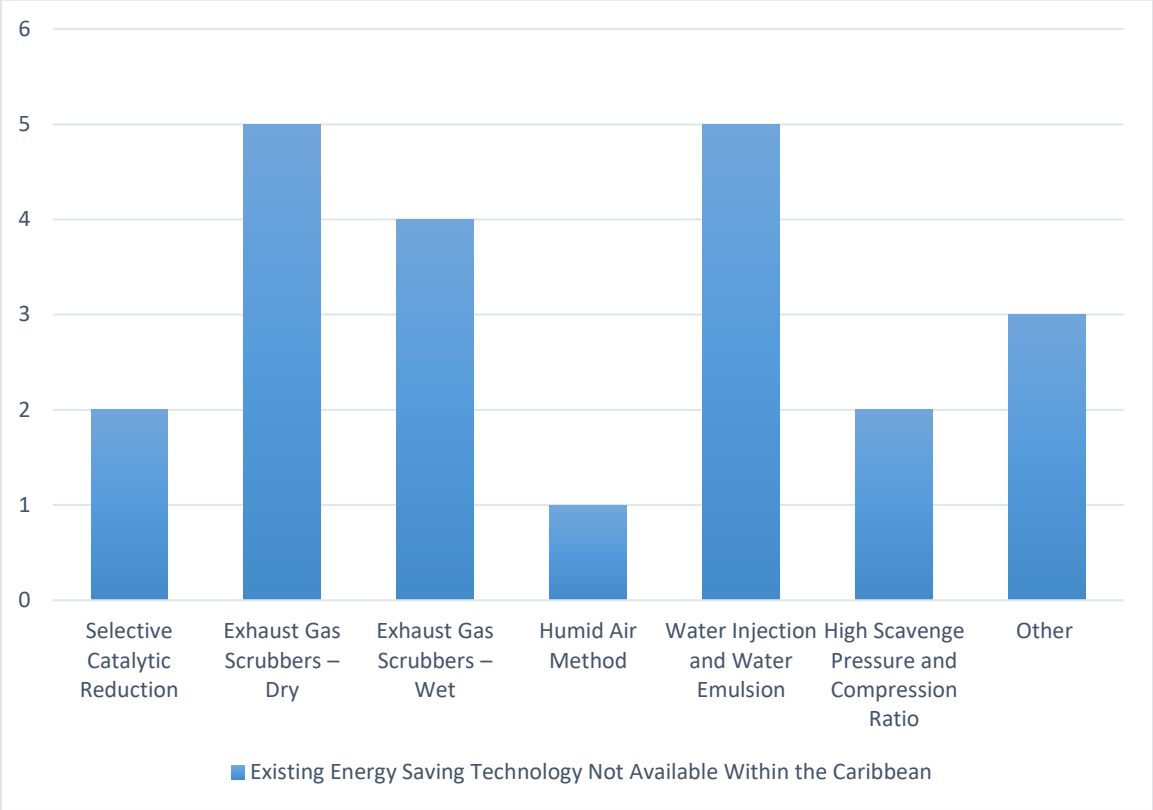
Three (3) of the eleven (11) respondents chose the “other” option and their responses were as follows:

- NIL
- NIL
- Battery Technology

10.12.1.2 Emission Abatement Technology

Eight (8) of the eleven (11) respondents identified emissions abatement technology that is not available within shipyards in the Caribbean region, which they would like installed/retrofitted on board their vessels. The technologies are illustrated in figure 18 below.

Figure 18 - Existing Emission Abatement Technology



Notably, the ship owners and operators identified each of the 6 emission abatement technologies as being unavailable within the Caribbean region.

Three (3) of the eleven (11) respondents chose the “other” option and their responses were as follows:

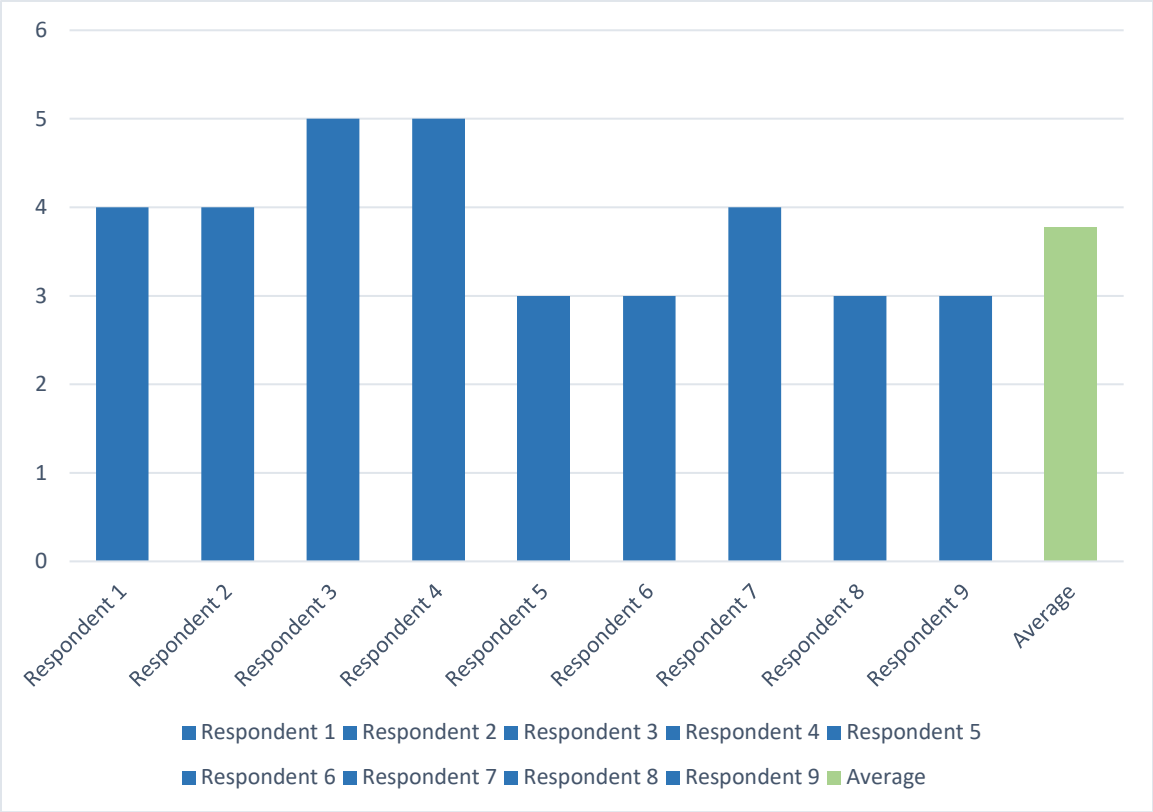
- NIL
- I am not sure
- LNG fueled propulsion system

10.12.2 Technology Gap

10.12.2.1 Awareness of Energy Efficient /Emission Abatement Technology

Nine (9) of the eleven (11) respondents indicated on a scale of 1 to 5, their level of awareness concerning Energy Efficient/Emission Abatement Technology, while two (2) respondents did not answer the question. The levels of awareness are illustrated in figure 19 below.

Figure 19 - Awareness of Energy Efficient /Emission Abatement Technology



The respondents displayed varying levels of awareness with the average awareness for owners/operators being 3.78 out of 5.

10.12.2.2 Gap between technology available within the Caribbean and internationally available energy efficient technologies

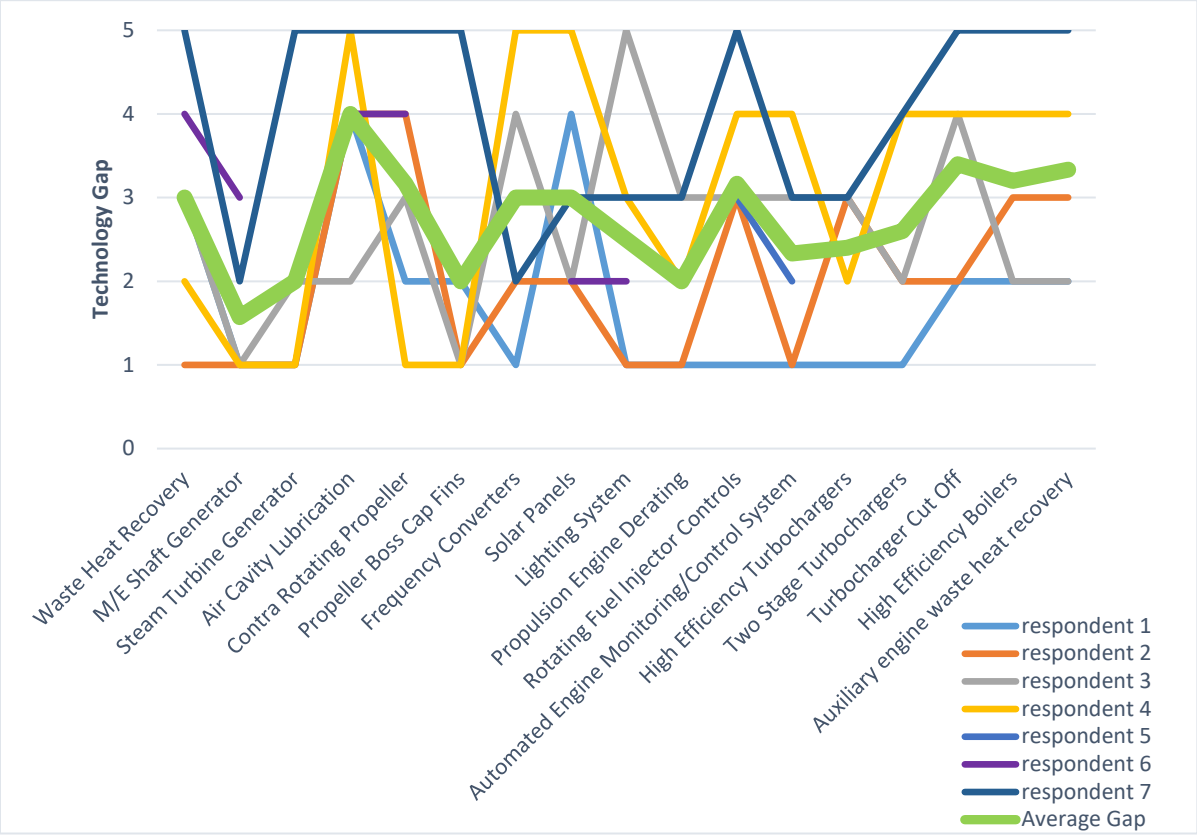
The respondents were asked to identify the gaps between energy efficient technology available within the Caribbean and energy efficient technologies available globally using the following rating scale;

- 1 - very small gap
- 2 - small gap
- 3 - medium gap
- 4 - large gap

- 5 - very large gap

Seven (7) of the eleven (11) respondents identified the gaps, while four (4) respondents did not answer the question. The technology gaps are illustrated in figure 21 below.

Figure 20 - Gap between technology available within the Caribbean and internationally available energy efficient technologies.

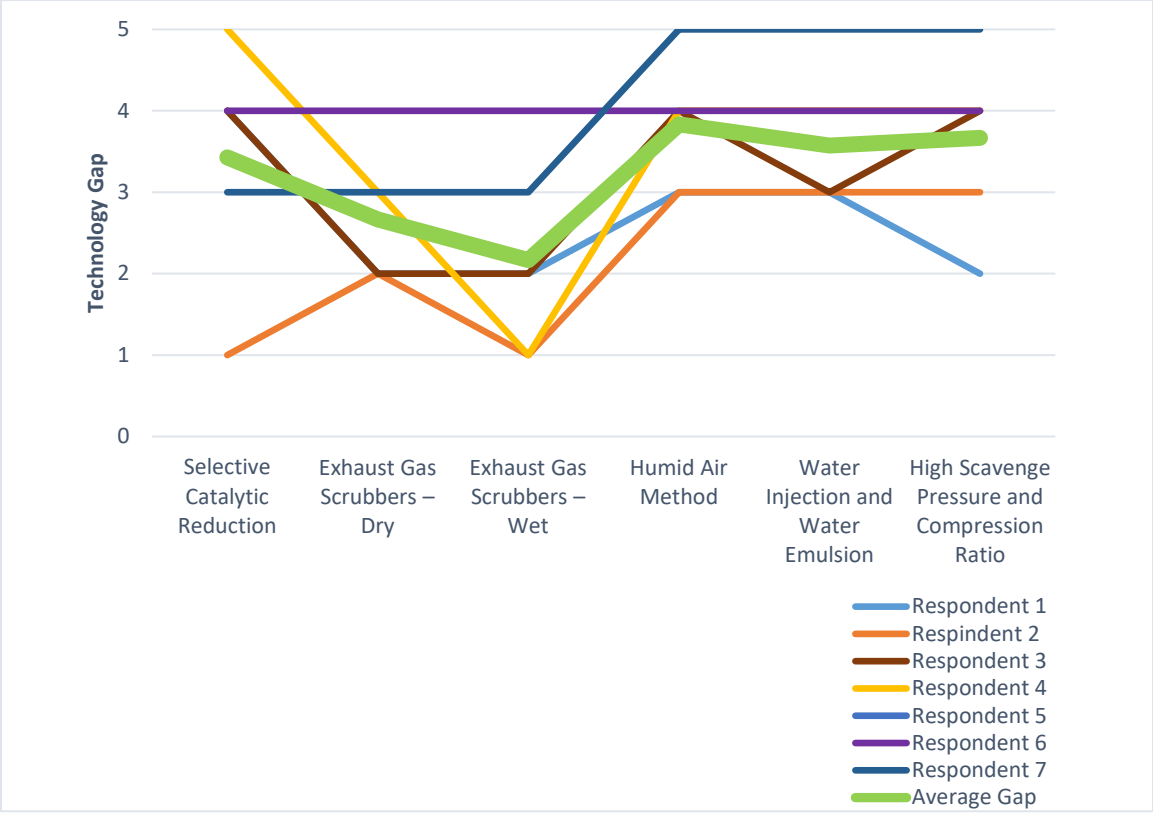


The gaps between energy efficient technologies available within the Caribbean and energy efficient technology available globally ranged from 1 to 5, varying widely among respondents. The average gap for all technologies identified by the respondents is 2.75, which is closer to a medium gap (3).

10.12.2.3 Gap between technology available within the Caribbean and internationally available emission abatement technology

The respondents were asked to identify the gaps between emission abatement technology available within the Caribbean and emission abatement technologies using the previously outlined gap rating scale. Seven (7) of the eleven (11) respondents identified the gaps, while four (4) respondents did not answer the question. The technology gaps are illustrated in figure 21 below.

Figure 21 - Gap between technology available within the Caribbean and internationally available emission abatement technologies.



The gaps between emission abatement technologies available within the Caribbean and emission abatement technology available globally ranged from 1 to 5, varying among respondents. The average gap for all technologies identified by the respondents is 3.22, which is closer to a medium gap (3).

10.12.3 Technology Needs

10.12.3.1 Factors considered in the determination of technology upgrades

Nine (9) of the eleven (11) respondents identified and ranked several factors in order of priority that apply when identifying the need for technological upgrades to vessels owned or operated by their company. Two (2) respondents did not answer the question. The factors are outlined in table 10 below.

Table 10 - Factors considered in the determination of technology upgrades

Factors	Respondent 1	Respondent 2	Respondent 3	Respondent 4	Respondent 5	Respondent 6	Respondent 7	Respondent 8	Respondent 9
Socio-Economic Factors	2		1	1	8		8		
Capacity (Insufficient existing technologies)	6		1		5		2		
Environmental Factors: GHG Reduction	7	1	1	2	4	1	5	2	
Environmental Factors: SOx Reduction	8	2	1	3	3	1	4	3	
Adaptation Measures	5		1		7		1		
Profitability	3	3	1		2		2	1	1
Increased Efficiency	4	3	1		6	1	3	4	2
Regulatory Compliance / Requirement	1	1	1		1	1	1		3

Six (6) of the nine (9) respondents prioritized Regulatory Compliance / Requirement as their top priority.

10.12.3.2 Factors considered when selecting technology to perform upgrades

Nine (9) of the eleven (11) respondents identified and ranked several factors in order of priority that apply when selecting the appropriate technology for vessels owned or operated by their company. Two (2) respondents did not answer the question. The factors are outlined in table11 below.

Table 11 - Factors considered when selecting technology to perform upgrades

Factors	Respondent 1	Respondent 2	Respondent 3	Respondent 4	Respondent 5	Respondent 6	Respondent 7	Respondent 8	Respondent 9
Emission Reduction Factors	5	1	1			1	5	3	
Return on Investment (ROI) (Cost Benefit Analysis)	2	1	1				1	2	1

Availability of Financial Support	4		1	1	3	1	1		3
Ease of Installation /Retrofit on existing vessels	3	2	1		4		1	1	
Capacity (Insufficient existing technologies)	6		1				1		
Financial Incentives	7	3	1	2			1		2
Upcoming Regulations/Future Legal Requirements	1	1	1	3	1	1	1	4	4
Availability within the Region	6	1	1		2		2	5	

Six (6) of the nine (9) respondents prioritized Upcoming Regulations/Future Legal Requirements as their top priority.

10.12.4 Barriers to the implementation of technology to meet technology needs

10.12.4.1 Barriers impacting the implementation of maritime technology transfer

The respondents were asked to rate the barriers affecting their company that impact the implementation of maritime technology transfer using the following rating scale;

- 0 – No impact
- 1 – Low Impact
- 2 – Moderate Impact
- 3 – Medium Impact
- 4 – Significant Impact
- 5 – Major Impact

Nine (9) of the eleven (11) respondents identified the barriers affecting maritime technology transfer and two (2) respondents did not answer the question. The barriers and their ratings are outlined in table 12 below;

Table 12 - Barriers impacting the implementation of maritime technology transfer

Factors	Respondent 1	Respondent 2	Respondent 3	Respondent 4	Respondent 5	Respondent 6	Respondent 7	Respondent 8

Lack of Finance	3	1	2	2		4	4	4
Lack of Incentives	1	2	0	2		4	3	4
Technical Feasibility	4	2	0	4		5	4	2
No Guarantee of Return on Investment	0	4	2	3		5	3	4
No regulatory requirements	5	4	4	5		5	4	4
High Investment Cost	5	2	2	4		5	3	2
Organizational Challenges	2	2	0	2	3	2	4	1
Non-Availability of Technology in Region	3	3	2	3		3	4	2
Lack of Public-Private Partnerships	1	0	1	1		4	4	3
Financial Incentive	1	5	1	3	3	5	4	4
Contractual Incentive	3	4	2	1	3	5	4	4

10.12.5 Solutions to the barriers to maritime technology transfer

The respondents were asked what they believe are the necessary actions to overcome the barriers to maritime technology transfer. Eight (8) of the eleven (11) respondents provided the following answers:

1. Financial incentives such as tax breaks or rebates.
2. Meeting regulatory compliance with GHG reduction is the major key for efficient operation.
3. To reduce commercial impact and the law must be clear without any grey area.
4. Make it regulatory requirement.
5. To be able to overcome barriers the company should be able to obtain contracts so to be able to have better financial incentives to overcome organizational challenges. This will aid in the company meeting all Maritime Technology transfers as needed.
6. Financial incentives to allow for midlife upgrades to existing tonnage and a ship building fund to allow for the purchase of suitable new tonnage.
7. Tried and tested, cost effective and value to all stakeholders.

Greater incentives should be made available to vessel owners to invest in new technologies.

10.13 Appendix XIII – Survey results: Technology Providers

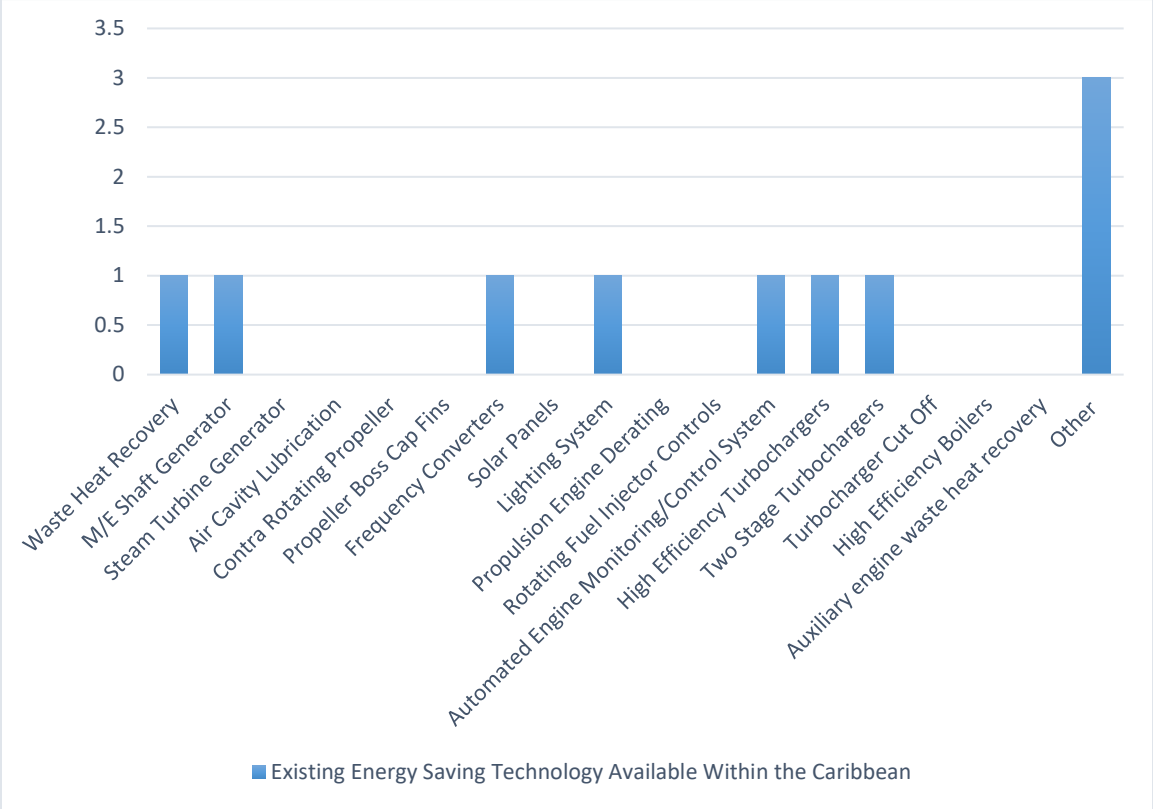
MTCC Caribbean received survey responses from five (5) technology providers, which are detailed below

10.13.1 Existing Technology

10.13.1.1 Energy Efficient Technology

Figure 22 below illustrates the existing technology available within the Caribbean region for installation/retrofit on board vessels.

Figure 22 - Existing Energy Efficient Technology Available Within the Caribbean



Of the 17 listed energy efficient technologies, the technology providers indicated that 7 are available within the Caribbean region for installation/retrofit on board vessels.

Notably, 3 of the 5 respondents chose the “other” option, and their responses were as follows;

6. Wind assist technology - Flettner rotors are already available. Other rigs are available on a case-by-case basis - bespoke rigs will be available, especially if testing rigs for wider installation in the fleet or fleet segment.

7. Energy Monitoring & Advisory systems, Remote Diagnostics, Energy Storage solutions, digital transformation, electric propulsion
8. Wind Energy (kites) Weather Routing System

10.13.1.2 Emission Abatement Technology

According to the technology providers, there are no Emission Abatement technologies available for retrofit/installment on board vessels in the Caribbean region.

10.13.1.3 Availability of Energy Efficient/ Emission Abatement technologies

The following locations were identified by the technology providers based on the availability of the energy efficient technologies previously listed;

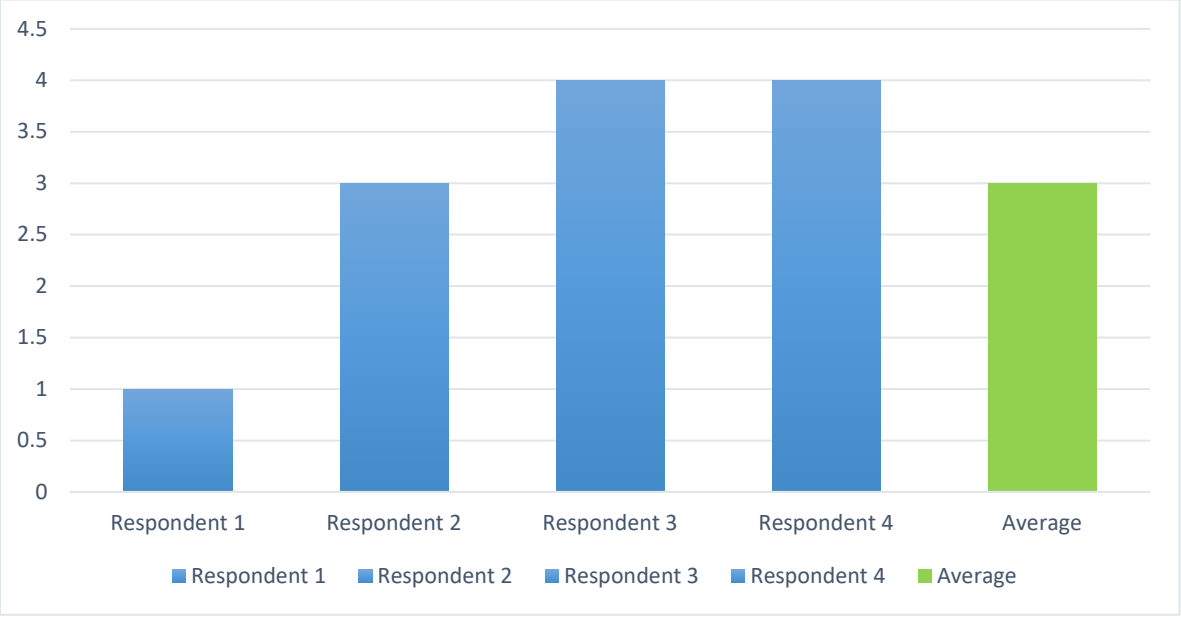
1. Global
2. We have over 30 technology providers for wind propulsion systems that are in various levels of market readiness - Flettner rotors are currently commercially available from two providers - Finland and UK, though those rigs can be delivered and installed internationally. Other rigs are primarily available from EU based companies, however there are a number under development in Asia and the US
3. The Energy Governance system is installed in hundred vessels all over the world

10.13.2 Technology Gap

10.13.2.1 Awareness of Energy Efficient /Emission Abatement Technology

Four (4) of the five (5) respondents indicated on a scale of 1 to 5, their level of awareness concerning Energy Efficient/Emission Abatement Technology, which is illustrated in figure 23 below.

Figure 23 - Awareness of Energy Efficient /Emission Abatement Technology



The respondents displayed varying levels of awareness with the average awareness for technology providers being 3 out of 5

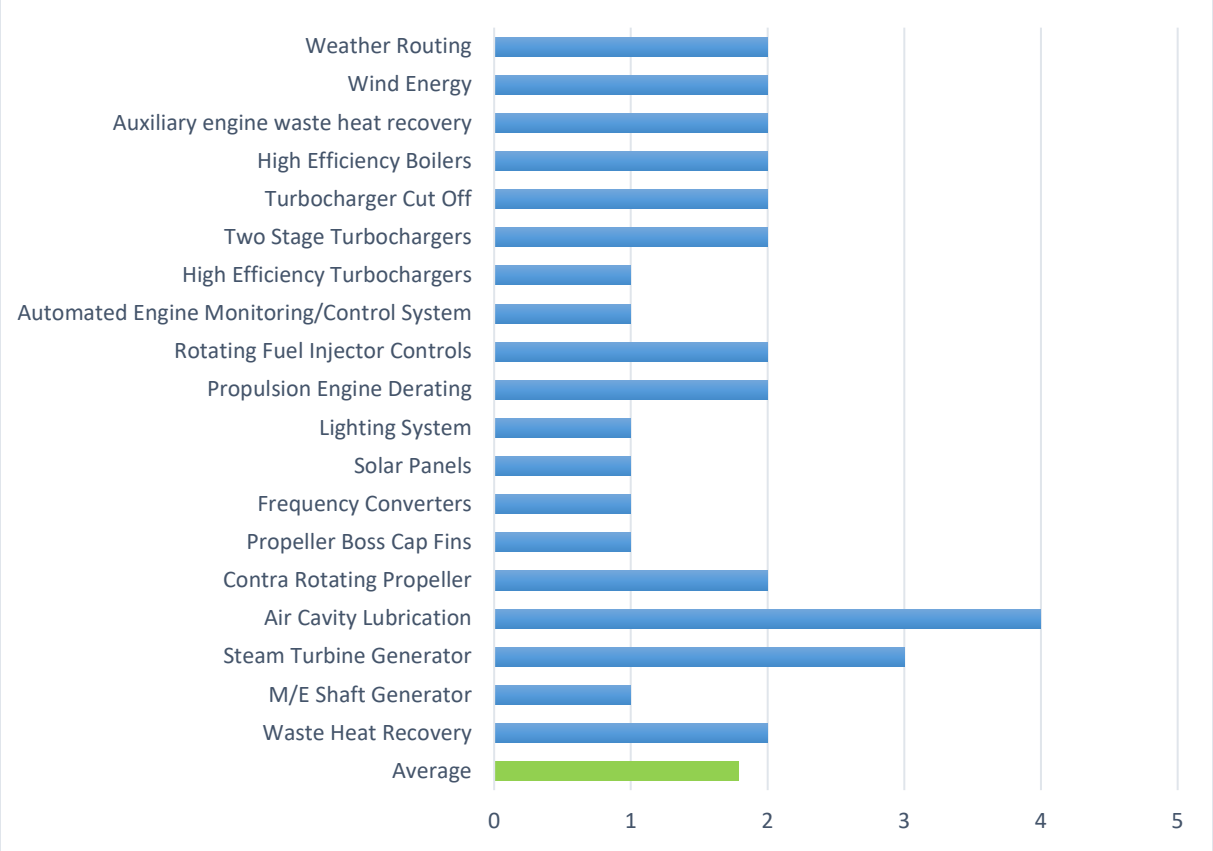
10.13.2.2 Gap between technology available within the Caribbean and internationally available energy efficient technologies

The respondents were asked to identify the gaps between energy efficient technology available within the Caribbean and energy efficient technologies available globally using the following rating scale;

- 1 - very small gap
- 2 - small gap
- 3 - medium gap
- 4 - large gap
- 5 - very large gap

One (1) of the five (5) respondents identified the gaps, another respondent clarified the question does not fall within their field of expertise and three (3) respondents did not answer the question. The technology gaps are illustrated in figure 24 below.

Figure 24 - Gap between technology available within the Caribbean and internationally available technologies.



The gaps between energy efficient technologies available within the Caribbean and energy efficient technology available globally ranged from 1 to 4, with 15 of the 17 technologies being rated either 1 (very small gap) or 2 (small gap). There was a medium gap for Steam Turbine Generation a large gap for Air Cavity Lubrication. The average technology gap between the Caribbean and globally available technology is 1.79, which is closer to a small gap (2).

10.13.2.3 Gap between technology available within the Caribbean and internationally available emission abatement technology

The respondents did not identify any emissions abatement technology, therefore no gaps were identified.

10.13.3 Technology Needs

10.13.3.1 Factors considered in the determination of technology upgrades

Two (2) of the five (5) respondents identified and ranked several factors in order of priority that apply when identifying the need for technological upgrades to vessels retrofitted by their company. One (1) respondent selected the "other" option, and two (2) respondents did not answer the question. The factors are outlined in table 13 below.

Table 13 - Factors considered in the determination of technology upgrades

Factors	**Respondent 1 priority ranking**	**Respondent 2 priority ranking**
Socio-Economic Factors	8	4
Capacity (Insufficient existing technologies)	5	8
Environmental Factors: GHG Reduction	4	2
Environmental Factors: SOx Reduction	7	3
Adaptation Measures	6	7
Profitability	2	1
Increased Efficiency	3	5
Regulatory Compliance / Requirement	1	6

Neither respondent rated any priority using the same rating.

Additionally, the respondent that selected the “other” option stated the following;

1. We don't fit vessels for equipment but the services we provide are driven by cost savings and mandatory compliance

10.13.3.2 Factors considered when selecting technology to perform upgrades

Two (2) of the five (5) respondents identified and ranked several factors in order of priority that apply when selecting the appropriate technology for vessels retrofitted by their company. One (1) respondent selected the “other” option, and two (2) respondents did not answer the question. The factors are outlined in table 14 below.

Table 14 - Factors considered when selecting technology to perform upgrades

Factors	**Respondent 1 priority ranking**	**Respondent 2 priority ranking**
Emission Reduction Factors	6	4
Return on Investment (ROI) (Cost Benefit Analysis)	1	1
Availability of Financial Support	2	2
Ease of Installation /Retrofit on existing vessels	5	5
Capacity (Insufficient existing technologies)	8	6
Financial Incentives	3	7
Upcoming Regulations/Future Legal Requirements	4	3
Availability within the Region	7	8

Notably, the top priorities of both respondents were “Return on Investment (ROI) (Cost Benefit Analysis)” and “Availability of Financial Support”.

Additionally, the respondent that selected the “other” option stated the following;

1. We don't fit vessels for equipment

10.13.4 Barriers to the implementation of technology to meet technology needs

10.13.4.1 Barriers impacting the implementation of maritime technology transfer

The respondents were asked to rate the barriers affecting their company that impact the implementation of maritime technology transfer using the following rating scale;

- 0 – No impact
- 1 – Low Impact
- 2 – Moderate Impact
- 3 – Medium Impact
- 4 – Significant Impact
- 5 – Major Impact

Three (3) of the five (5) respondents identified the barriers affecting maritime technology transfer and Two (2) respondents did not answer the question. The barriers are outlined in table 15 below.

Table 15 - Barriers impacting the implementation of maritime technology transfer

Factors	**Respondent 1 impact rating**	**Respondent 2 impact rating**	**Respondent 3 impact rating**
Lack of Finance	5	5	5
Lack of Incentives	4	4	2
Technical Feasibility	5	1	2
No Guarantee of Return on Investment	5	4	4
No regulatory requirements	3	4	4
High Investment Cost	5	4	5
Organizational Challenges	4	1	1
Non-Availability of Technology in Region	2	2	0
Lack of Public-Private Partnerships	2	1	2

Class Approvals	4	1	0
Intellectual Property Protection	3	1	1
Access to Trials on Ships	3	3	2

Respondent two, whom noted in the technology needs section that their company does not fit vessels for equipment, also noted that the identified barriers and their impacts were selected in the context of providing services to shipping companies or national agencies.

10.13.5 Solutions to the barriers to maritime technology transfer

The respondents were asked what they believe are the necessary actions to overcome the barriers to maritime technology transfer and the following was stated by Two (2) of the five (5) respondents.

8. Increasing fuel prices will lead for more openness to investing new technologies
9. Lack of meaningful data (right data, right place, right time), lack of financial incentives, lack of funding for national agencies to be able to put in place the infrastructure required to provide the regulation and support necessary

10.14 Appendix XIV – Survey results: Ports

MTCC Caribbean received survey responses from four (4) Ports and Port Authorities, which are detailed below

10.14.1 Existing Technology

10.14.1.1 Energy Efficient Technology

Each of the four (4) respondents indicated that there is no energy efficient technology at their facilities that is available for use by vessels calling at their port.

10.14.1.2 Emission Abatement Technology

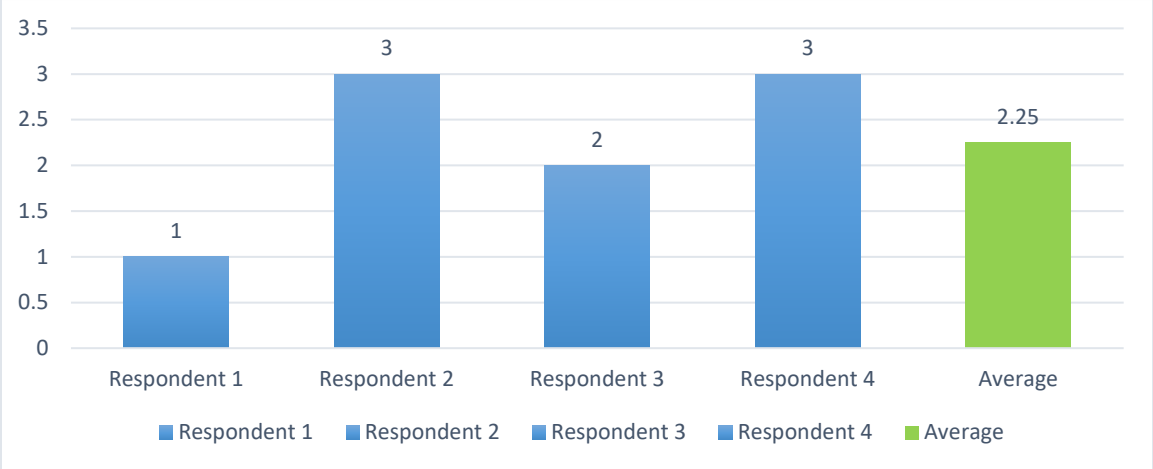
Each of the four (4) respondents indicated that there is no emission abatement technology at their facilities that is available for use by vessels calling at their port.

10.14.2 Technology Gap

10.14.2.1 Awareness of Energy Efficient /Emission Abatement Technology

The four (4) respondents indicated on a scale of 1 to 5, their level of awareness concerning Energy Efficient/Emission Abatement Technology, which is illustrated in figure 25 below.

Figure 25 - Awareness of Energy Efficient /Emission Abatement Technology



The respondents displayed varying levels of awareness with the average awareness for Ports and Port Authorities being 2.25 out of 5.

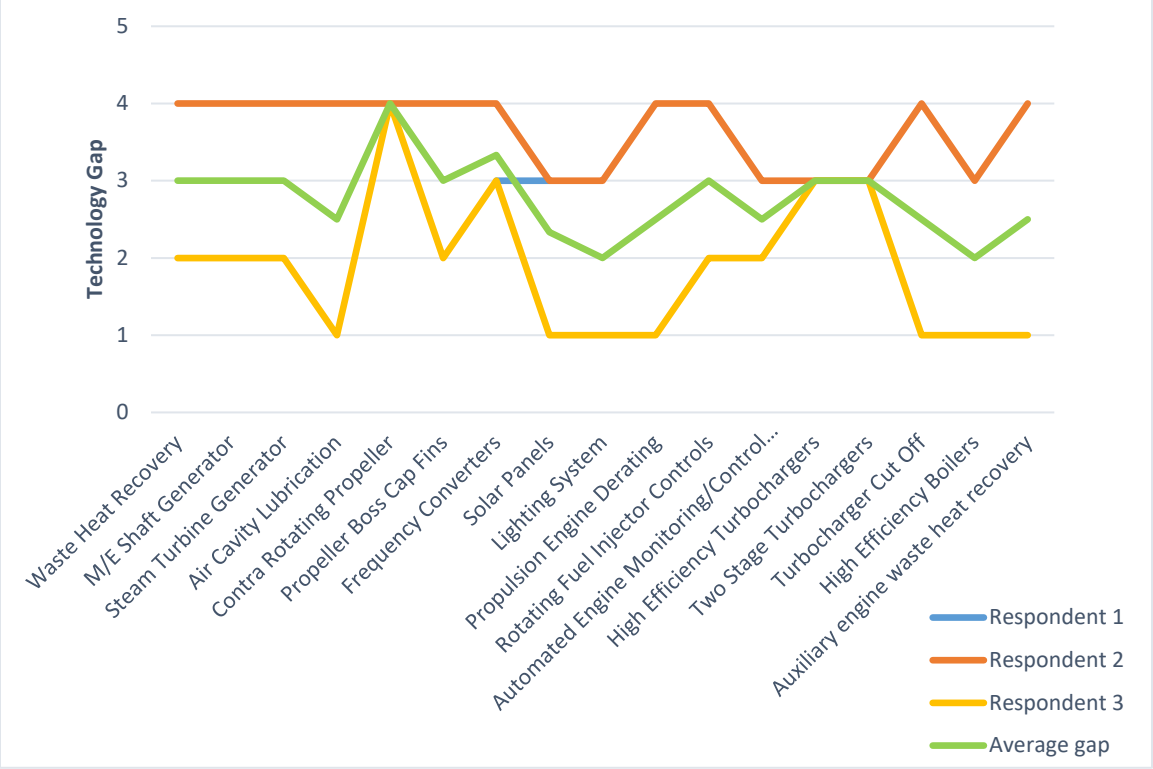
10.14.2.2 Gap between technology available within the Caribbean and internationally available energy efficient technologies

The respondents were asked to identify the gaps between energy efficient technology available within the Caribbean and energy efficient technologies available globally using the following rating scale;

- 1 - very small gap
- 2 - small gap
- 3 - medium gap
- 4 - large gap
- 5 - very large gap

Three (3) of the four (4) respondents identified the gaps, while the fourth respondent did not answer the question. The technology gaps are illustrated in figure 26 below.

Figure 26 - Gap between technology available within the Caribbean and internationally available energy efficient technologies.

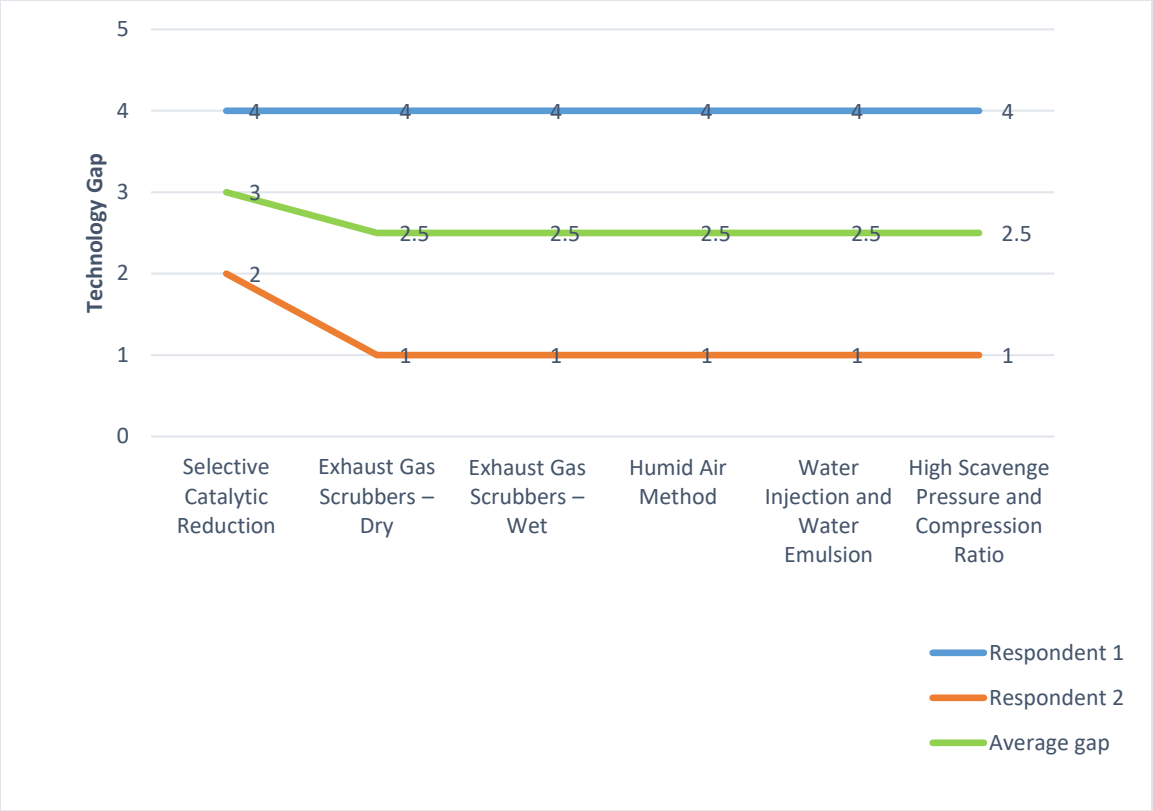


The gaps between energy efficient technologies available within the Caribbean and energy efficient technology available globally ranged from 1 to 4, varying widely among respondents. Notably, one respondent identified gaps for just 2 of the 17 technologies. The average gap for all technologies identified by the respondents is 2.77, which is closer to a medium gap (3).

10.14.2.3 Gap between technology available within the Caribbean and internationally available emission abatement technology

The respondents were asked to identify the gaps between emission abatement technology available within the Caribbean and emission abatement technologies using the previously outlined gap rating scale. Two (2) of the four (4) respondents identified the gaps, while two (2) respondents did not answer the question. The technology gaps are illustrated in figure 27 below.

Figure 27 - Gap between technology available within the Caribbean and internationally available emission abatement technologies.



The gaps between emission abatement technologies available within the Caribbean and emission abatement technology available globally ranged from 1 to 4, varying widely between both respondents. The average gap for all technologies identified by the respondents is 2.58, which lies roughly between a small gap (2) and a medium gap (3).

10.14.3 Technology Needs

10.14.3.1 Factors considered in the determination of technology upgrades

Two (2) of the four (4) respondents identified and ranked several factors in order of priority that apply when identifying the need for technological upgrades to vessels operating within their organization's jurisdiction. Two (2) respondents did not answer the question. The factors are outlined in table 16 below.

Table 16 - Factors considered in the determination of technology upgrades

Factors	**Respondent 1 priority ranking**	**Respondent 2 priority ranking**
Socio-Economic Factors	8	2
Capacity (Insufficient existing technologies)	1	1
Environmental Factors: GHG Reduction	7	
Environmental Factors: SOx Reduction	5	
Adaptation Measures	6	5
Profitability	2	
Increased Efficiency	3	3
Regulatory Compliance / Requirement	4	4

Both respondents shared priority rankings for three factors; Capacity (Insufficient existing technologies), Increased Efficiency and Regulatory Compliance / Requirement, ranking them as their 1st, 3rd and 4th priorities respectively.

10.14.3.2 Factors considered when selecting technology to perform upgrades

Two (2) of the four (4) respondents identified and ranked several factors in order of priority that apply when selecting the appropriate technology for vessels operating within their organization's jurisdiction. Two (2) respondents did not answer the question. The factors are outlined in table 17 below.

Table 17 - Factors considered when selecting technology to perform upgrades

Factors	**Respondent 1 priority ranking**	**Respondent 2 priority ranking**
Emission Reduction Factors		5
Return on Investment (ROI) (Cost Benefit Analysis)		2
Availability of Financial Support		1
Ease of Installation /Retrofit on existing vessels	4	
Capacity (Insufficient existing technologies)	1	3
Financial Incentives		6
Upcoming Regulations/Future Legal Requirements	2	4
Availability within the Region	3	7

There were no similarities in priority ranking between both respondents when considering factors that affect the selection of technology to perform upgrades.

10.14.4 Barriers to the implementation of technology to meet technology needs

10.14.4.1 Barriers impacting the implementation of maritime technology transfer

The respondents were asked to rate the barriers affecting their organisation that impact the implementation of maritime technology transfer using the following rating scale;

- 0 – No impact
- 1 – Low Impact
- 2 – Moderate Impact
- 3 – Medium Impact
- 4 – Significant Impact
- 5 – Major Impact

Two (2) of the four (4) respondents identified the barriers affecting maritime technology transfer and Two (2) respondents did not answer the question. The barriers are outlined in table 17 below.

Table 17 - Barriers impacting the implementation of maritime technology transfer

Factors	**Respondent 1 impact rating**	**Respondent 2 impact rating**
Lack of Finance	4	5
Lack of Incentives	4	4
Technical Feasibility	3	3
No Guarantee of Return on Investment	3	4
No regulatory requirements	1	4
High Investment Cost	5	5
Organizational Challenges	3	2
Non-Availability of Technology in Region	4	2
Lack of Public-Private Partnerships	4	4

There were similarities between the levels of impact identified for the barriers by both respondents.

10.14.5 Solutions to the barriers to maritime technology transfer

The respondents were asked what they believe are the necessary actions to overcome the barriers to maritime technology transfer and the following was stated by Two (2) of the four (4) respondents.

11. A more practical approach is needed in order to meet climate change needs and the implementation of energy saving / green technology
12. Awareness of positive effect on the bottom line of a company or country by extension